
Exact Depth Scaling and Low-Rank Feynman Calculus for Multi-Component Multilayer Networks

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Abstract

The depth exponents displayed in recent finite-width neural-tangent-kernel (NTK) Feynman plots were obtained by Monte Carlo regression, and their noninteger values were not exact critical powers. We close this gap for fully connected critical ReLU networks and extend the diagrammatics to multi-component and multilayer neural networks (MMNNs), whose blocks have the form $h_\ell = A_\ell \text{ReLU}(W_\ell h_{\ell-1})$, with frozen W_ℓ and trainable A_ℓ .

First, we derive a pathwise matrix-valued NTK recursion for the actual MMNN, without a Gaussian-process or infinite-rank approximation. Concatenating two blocks inserts the rank- r tangent metric $G = XX^\top/r$. Its moments and connected cumulants are computed at every valence by a closed Wick-pairing formula. A QR decomposition separates this vertex into an independent Wishart radial part and a Haar–Stiefel orientation; orthogonal Weingarten calculus gives an exact all-valence formula for the latter. This distinction corrects the common projector-only replacement, which is exact only after whitening the induced parameter metric.

Second, working directly with the connected-cumulant definitions of Guillen–Misof–Gerken, we solve the deterministic leading-width ReLU recursions for the five non-MMNN four-index tensors V, D, F, A, B . For every fixed non-collinear input set, all components plotted in Guillen–Misof–Gerken have the attained powers

$$V = \Theta(L), \quad D, F = \Theta(L^2), \quad A, B = \Theta(L^3).$$

We give every leading coefficient in closed form, prove non-cancellation, and derive the logarithmic critical drift $\theta_{ab,L} = 3\pi/[L + (3/2)\log L + C_{ab} + o(1)]$. Applying the original regression windows to our exact recursion explains the reported effective slopes between 1.19 and 3.78 without introducing new anomalous powers.

Third, we lift these five cumulants to the MMNN component indices. Their diagrams split into a width-colored sector and a bottleneck-rank-colored sector. In the controlled double expansion, their attained depth powers are again 1, 2, 2, 3, 3, while their amplitudes scale as $1/n + 1/r$ for Gaussian MMNNs and as $1/n + \gamma_{n,r}$ for the whitened Stiefel model. We give the exact contraction formula for each rank coefficient and delimit the perturbative depth window.

Fourth, we show that the complete MMNN tangent feature is an exact concatenation of low-rank Kronecker-Gram blocks. Under the Benigni–Paquette quadratic-scaling hypotheses, its spectrum is one Marchenko–Pastur map of the full block population covariance. For homogeneous decoupled blocks we obtain the closed law $\mu_{\text{MP}}^{\gamma_1/L} \boxtimes (L\cdot)_{\#} \mu_{\nu,\varphi}$ and its exact linear spectral response.

Fifth, we connect the diagrammatics to the symmetry-derived neural quadratic form (NQF) of Liu Ziyin, Xu, Poggio, and Chuang. We identify its structure matrix with a two-leg Landau vertex and $M = WW^\top$ with a dynamic Gram order parameter. All chains of quadratic vertices close exactly on M ; in the commuting orthogonal-feature sector this gives the complete finite-sample NTK spectrum and logistic mode ignition. Combining the exact Wishart and Haar–Stiefel seed laws

with this flow yields explicit finite-rank ignition-time distributions, including exact cancellation of clock disorder at the full-rank Stiefel endpoint. We state precisely why this is an exact resummation for the NQF but only a local normal form for a nonlinear MMNN.

Finally, the finite-size tensor covariance is resummed spectrally. In an isotropic large-rank sector, its connected-cumulant hierarchy yields a deformed-Wigner Dyson equation. We compute limiting trace moments, eigenvalue density, and basis-resolved eigenvector overlaps, and insert the resulting local spectral measures into power-law random-feature learning dynamics. A positive-semidefinite deformation shows that finite-rank mixing need not always move early stopping earlier: the direction is determined by the full diagrammatic covariance. All symbolic identities, deterministic recursions, spectral solvers, and figures are accompanied by executable tests.

1 Introduction

Finite-width neural networks are perturbations of their infinite-width kernels, but the perturbation is structured. Its entries share neurons, gates, backpropagated paths, and—in a low-rank network—the same random subspaces. Entrywise central-limit intuition is therefore insufficient for depth scaling or spectral questions. Feynman diagrams provide the right bookkeeping: channel loops determine powers of width or rank, while connected vertices determine which tensor cumulants survive.

Dyer and Gur-Ari established the large-width diagrammatic power counting and related derivative-tensor contractions to spectral moments [3]. Those relations do not imply a universal Wigner law for nonlinear NTKs: the covariance sector must first be computed. Our spectral analysis therefore uses their moment viewpoint but closes it with a covariance-specific matrix Dyson equation.

Two recent developments make a sharper theory possible. Guillen, Misof, and Gerken derive exact recursive expressions for the leading finite-width tensors of deep nonlinear networks and report componentwise depth slopes from finite-network Monte Carlo experiments [4]. Separately, MMNNs compose shallow random-feature maps $A_\ell \sigma(W_\ell \cdot + b_\ell) + c_\ell$, training the component matrices A_ℓ, c_ℓ while freezing W_ℓ, b_ℓ [5]. The adjacent product $W_{\ell+1} A_\ell$ is a low-rank effective weight with a frozen left and trainable right factor. The natural question is whether the dense Feynman tensor theory can be made exact in depth and extended to this alternating width–rank architecture.

Benigni and Paquette compute an asymptotic eigenvalue distribution for a two-layer NTK in the quadratic sample–parameter scaling [6]. That result is complementary spectral input; it does not by itself supply a multilayer concatenation identity. Here the concatenation rule instead follows pathwise from the chain rule.

The first answer is a correction: the slopes printed in the depth plots are not exact exponents. They are least-squares slopes of empirical finite-width tensors over depths at most 30. At critical ReLU initialization, distinct inputs approach perfect correlation only as L^{-2} , with a logarithmic drift. Consequently a tensor with true leading behavior cL^p can have a substantial effective slope different from p over this window. We evaluate the exact deterministic recursions and prove that the five powers are the integers 1, 2, 2, 3, 3, component by component.

The second answer is architectural. A Gaussian frozen factor is not a Haar projector. Writing $X = QR$ gives

$$\frac{1}{r} X X^\top = Q \frac{RR^\top}{r} Q^\top.$$

The orientation Q is Haar–Stiefel and the radial matrix $RR^\top$ is Wishart, independently. Dropping the radial vertex changes finite-rank moments and the Euclidean tangent metric. We therefore give parallel exact rules for (i) the Gaussian Gram vertex of the actual MMNN and (ii) the whitened Stiefel-projector surrogate. The former has connected k -vertex cumulants of order r^{1-k} ; the latter has finite- (n, r) rational Weingarten weights and an exact identity endpoint at $r = n$.

The third answer concerns eigenvalues. Weingarten calculus determines Haar orientation averages and eigenvector moments, but it does not by itself diagonalize a nonlinear NTK. The tensor recursion first supplies the covariance operator of the NTK correction. A matrix Dyson equation then resums

the leading spectral diagrams. In the isotropic case this reduces to the Pastur equation for a deformed Wigner matrix; in the general correlated case it is an operator-valued Dyson equation. This yields eigenvalue density and eigenvector overlaps from the same tensor data.

In the quadratic sample–parameter regime there is a second, nonperturbative base law. The Benigni–Paquette theorem applies to one Kronecker tangent block. The exact pathwise concatenation shows that a deep MMNN is the Gram matrix of all such transported blocks at once. A single MP map therefore acts on their full block population covariance; the off-diagonal layer blocks determine whether the simpler homogeneous formula is valid.

A third simplification occurs near a symmetry-fixed parameter origin. Liu Ziyin, Xu, Poggio, and Chuang prove that permutation symmetry together with their zero-gradient-at-zero condition forces the leading nonconstant term to be a quadratic form $\text{Tr}(WW^\top \mathcal{A}_x)$ [8]. This is a Landau normal form, not a claim that an arbitrary network is globally quadratic. In diagrammatic language $\mathcal{A}_x$ is a deterministic two-leg vertex. Because its gradient is linear in W , every repeated insertion closes on the dynamic Gram matrix $M = WW^\top$. This produces exact training-time resummations inside the NQF and a controlled local sector to be matched to the finite-width and finite-rank initialization diagrams developed here.

Main results.

1. We prove the exact pathwise MMNN NTK recursion, Theorem 2.1, and identify the concatenated low-rank tangent metric without changing the architecture.
2. We derive all-valence Gaussian Gram moment/cumulant rules, Theorem 7.1, and all-valence Haar-projector Weingarten rules, Theorem 7.2.
3. We give deterministic exact finite-depth ReLU recursions for V, D, F, A, B , including closed ReLU transport Hessians and a deterministic Tallis–Plackett evaluation of every source moment.
4. We prove attained componentwise powers and closed leading coefficients, Theorem 5.2. In particular none of the thirty plotted components cancels at leading order.
5. We compute the same finite-window regressions used in the original plots and separate them from the exact asymptotic powers.
6. We define the MMNN analogues of the same five connected cumulants and prove their two-color depth hierarchy, Theorem 8.1.
7. We prove the exact tangent-feature concatenation and, under the stated quadratic-scaling concentration hypotheses, derive the multilayer Benigni–Paquette law and its spectral response, Theorem 9.2 and Corollary 9.3.
8. We give an NQF–MMNN dictionary, prove the exact quadratic-vertex resummation and dynamic NTK formula, and compute the full spectrum and finite-rank mode-ignition clocks in the orthogonal-feature sector, Theorems 10.2, 10.4, and 10.5.
9. We derive spectral moments, the Dyson resolvent, eigenvalue density, and local eigenvector overlaps, and apply them to power-law scaling and early stopping.

Scope. The closed rational coefficients in Theorem 5.2 concern the leading $1/n$ coefficient of the dense critical ReLU expansion. The MMNN Feynman rules are exact at finite width and rank, while its exponent theorem is a controlled joint $1/n, 1/r$ expansion; a fixed-rank, arbitrarily deep limit is outside that expansion. The quadratic-scaling law additionally requires convergence and quadratic-form concentration of the transported population covariance; its direct application to the discontinuous ReLU derivative requires a universality extension. The Wigner reduction in Section 11 requires a separate large-data isotropy assumption. We distinguish exact identities, asymptotic theorems, and conditional random-matrix reductions throughout. Nonnegative matrix factorization constraints are not present in the MMNN architecture studied in [5]; adding positivity would replace the tangent space by an active cone and requires a separate theory. The NQF statements add a second, independent scope condition: closure is exact for the quadratic model, whereas transferring it to the original nonlinear network is local in parameter norm. Moreover, a single MMNN block trained only through its right factor lies in the linear part of the general normal form; a pure M-sector first appears for a bilinear pair of trainable blocks or for a layerwise deep NQF.

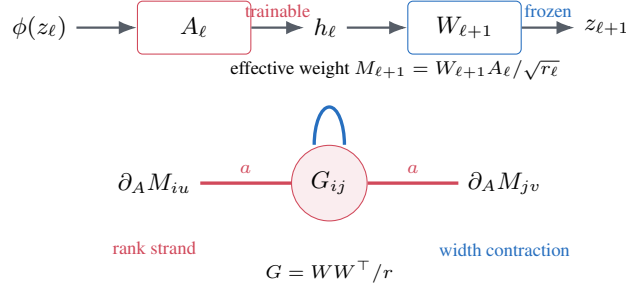


Figure 1: Concatenating two MMNN blocks produces a frozen-left/trainable-right effective weight. Contracting two trainable-right tangents creates the normalized Gram vertex G , whose component strand is red and frozen-feature contraction is blue. Gaussian Wick or Stiefel Weingarten rules evaluate this same vertex without changing the network.

2 The actual MMNN and its pathwise NTK

Let r_0 be the input dimension, r_L the output dimension, and n_ℓ the random-feature width of block ℓ . We use the normalized bias-free model

$$z_\ell(x) = \frac{1}{\sqrt{r_{\ell-1}}} W_\ell h_{\ell-1}(x), \quad W_\ell \in \mathbb{R}^{n_\ell \times r_{\ell-1}}, \quad (1)$$

$$h_\ell(x) = \frac{1}{\sqrt{n_\ell}} A_\ell \phi(z_\ell(x)) + c_\ell, \quad A_\ell \in \mathbb{R}^{r_\ell \times n_\ell}. \quad (2)$$

The frozen variables are W_ℓ ; the trainable variables are A_ℓ, c_ℓ . Scale constants can be absorbed into W, A and are restored when needed. Define

$$q_\ell(x, y) = \frac{1}{n_\ell} \phi(z_\ell(x))^\top \phi(z_\ell(y)) + 1, \quad (3)$$

$$J_\ell(x) = \frac{1}{\sqrt{n_\ell r_{\ell-1}}} A_\ell D_\ell(x) W_\ell, \quad D_\ell(x) = \text{diag } \phi'(z_\ell(x)). \quad (4)$$

Theorem 2.1 (Exact pathwise MMNN NTK recursion). *For every realization of the weights, every finite width and rank, and all inputs x, y , the matrix-valued NTK with respect to $\{A_s, c_s : s \leq \ell\}$ satisfies*

$$\Theta_\ell(x, y) = q_\ell(x, y) I_{r_\ell} + J_\ell(x) \Theta_{\ell-1}(x, y) J_\ell(y)^\top, \quad \Theta_0 = 0. \quad (5)$$

No independence, Gaussian-process, or infinite-width limit is used.

Equation (5) is the base object for MMNN diagrams. A direct vertex is born when A_ℓ, c_ℓ are differentiated; earlier vertices are transported by the random Jacobian J_ℓ . It also makes the low-rank metric precise. Combining adjacent blocks produces the effective matrix

$$M_{\ell+1} = \frac{1}{\sqrt{r_\ell}} W_{\ell+1} A_\ell. \quad (6)$$

For fixed right indices u, v , contraction of two derivatives with respect to A_ℓ gives

$$\sum_{a,k} \frac{\partial M_{iu}}{\partial (A_\ell)_{ak}} \frac{\partial M_{jv}}{\partial (A_\ell)_{ak}} = \delta_{uv} G_{ij}, \quad G = \frac{1}{r_\ell} W_{\ell+1} W_{\ell+1}^\top. \quad (7)$$

Thus the correct low-rank line is a Gram matrix. If $W = X$ has iid standard Gaussian entries, it is normalized Wishart. If $W = \sqrt{n} U$ with $U^\top U = I_r$, it is the normalized projector $G = (n/r) U U^\top$. These ensembles agree in mean but not in higher moments.

3 Feynman rules: width loops, rank loops, and connected vertices

The alternating indices of an MMNN have two colors. Greek feature indices $\mu \in [n]$ produce width loops; Latin component indices $a \in [r]$ produce rank loops. With iid Gaussian frozen factors,

$$\mathbb{E} W_{\mu a} W_{\nu b} = \delta_{\mu\nu} \delta_{ab}. \quad (8)$$

Every Wick contraction therefore closes both a width and a rank strand. A connected cumulant of k empirical feature vertices costs n^{1-k} , while a connected cumulant of k Gram vertices costs r^{1-k} . A connected MMNN diagram Γ consequently carries

$$\text{wt}(\Gamma) = n^{-\sum_v (k_v^{(n)} - 1)} r^{-\sum_w (k_w^{(r)} - 1)} \times \text{Contr}(\Gamma), \quad (9)$$

where $\text{Contr}(\Gamma)$ contains ReLU source and transport tensors. Equation (9) is exact before summing over layer locations. Depth powers arise from the number and transport of allowed vertex insertions.

At first connected order, an actual Gaussian MMNN has width and rank sources of orders $1/n$ and $1/r$; mixed diagrams can correlate their contractions. In an isotropic sector this gives the perturbative scale

$$\varepsilon_{n,r} = \frac{c_n}{n} + \frac{c_r}{r}, \quad (10)$$

with contractions c_n, c_r fixed by the diagram. It is generally wrong to replace (10) by a single projector defect. For the whitened Stiefel surrogate the orientation covariance instead contains the exact defect

$$\gamma_{n,r} = \frac{n(n-r)}{r(n-1)(n+2)}, \quad (11)$$

which vanishes at $r = n$. The ordinary width contribution remains.

4 Non-MMNN theory in the exact tensor framework

This section contains no MMNN or low-rank assumption. It uses exactly the neural-index decomposition of Guillen–Misof–Gerken [4]. If $\Delta\hat{\Theta} = \hat{\Theta} - \mathbb{E}\hat{\Theta}$, their five tensors are defined by

$$\text{cum}(z_{i_1;1}, z_{i_2;2}, z_{i_3;3}, z_{i_4;4}) = \frac{1}{n_{\ell-1}} [\delta_{i_1 i_2} \delta_{i_3 i_4} V_{1234} + \delta_{i_1 i_3} \delta_{i_2 i_4} V_{1324} + \delta_{i_1 i_4} \delta_{i_2 i_3} V_{1423}], \quad (12)$$

$$\text{cum}(z_{i_1;1}, z_{i_2;2}, \Delta\hat{\Theta}_{i_3 i_4;34}) = \frac{1}{n_{\ell-1}} [\delta_{i_1 i_2} \delta_{i_3 i_4} D_{1234} + \delta_{i_1 i_3} \delta_{i_2 i_4} F_{1324} + \delta_{i_1 i_4} \delta_{i_2 i_3} F_{1423}], \quad (13)$$

$$\text{cum}(\Delta\hat{\Theta}_{i_1 i_2;12}, \Delta\hat{\Theta}_{i_3 i_4;34}) = \frac{1}{n_{\ell-1}} [\delta_{i_1 i_2} \delta_{i_3 i_4} A_{1234} + \delta_{i_1 i_3} \delta_{i_2 i_4} B_{1324} + \delta_{i_1 i_4} \delta_{i_2 i_3} B_{1423}]. \quad (14)$$

Numerals abbreviate sample arguments $x_1, \dots, x_4$, never neural indices. Thus V is the connected four-preactivation coefficient, D, F are the two neural-index sectors of the mixed preactivation–NTK cumulant, and A, B are the two sectors of the NTK covariance. Equations (12)–(14) also fix all permutations and normalizations used below.

We now solve these non-MMNN leading-width coefficients. Let $g = (g_1, \dots, g_m) \sim \mathcal{N}(0, K)$, $R_a = (g_a)_+$, and $I_a = \mathbf{1}_{g_a > 0}$. At criticality $C_W = 2$, define

$$Q_{ab} = \mathbb{E} R_a R_b, \quad \dot{Q}_{ab} = \mathbb{E} I_a I_b, \quad (15)$$

$$\Omega_{ab} = R_a R_b + 2H_{ab} I_a I_b, \quad (16)$$

$$R_{ab}^{gh} = \mathbb{E} \partial_g \partial_h (R_a R_b), \quad O_{ab}^{gh} = \mathbb{E} \partial_g \partial_h \Omega_{ab}, \quad (17)$$

$$U_{ab}^g = \mathbb{E} \partial_g (R_a I_b). \quad (18)$$

The infinite kernels obey

$$K^+ = 2Q, \quad H^+ = Q + 2\dot{Q} \odot H. \quad (19)$$

For example, if $a \neq b$, writing $\theta_{ab} = \arccos(K_{ab}/\sqrt{K_{aa}K_{bb}})$,

$$Q_{ab} = \frac{\sqrt{K_{aa}K_{bb}}}{2\pi} [\sin \theta_{ab} + (\pi - \theta_{ab}) \cos \theta_{ab}], \quad (20)$$

$$\dot{Q}_{ab} = \frac{\pi - \theta_{ab}}{2\pi}. \quad (21)$$

All entries of R, O, U are closed elementary functions. The apparent singularity $\mathbb{E}[\delta(g_a)\delta(g_b)] \sim 1/\sin \theta_{ab}$ is essential: it amplifies transverse collision sectors in D and A .

Let $C_{RR,RR} = \text{Cov}(R_a R_b, R_c R_d)$, and define $C_{RR,\Omega}$, $C_{\Omega,\Omega}$ analogously. With repeated neural indices summed, the exact layer update implemented in our calculations is

$$V_{abcd}^+ = 4C_{RR,RR}^{ab;cd} + R_{ab}^{gh} V_{ghij} R_{cd}^{ij}, \quad (22)$$

$$D_{abcd}^+ = 2C_{RR,\Omega}^{ab;cd} + \frac{1}{2} R_{ab}^{gh} V_{ghij} O_{cd}^{ij} + 2\dot{Q}_{cd} R_{ab}^{gh} D_{ghcd}, \quad (23)$$

$$F_{abcd}^+ = 4\mathbb{E}[R_a R_c I_b I_d] H_{bd} + 4U_{ab}^g U_{cd}^h F_{ghbd}, \quad (24)$$

$$A_{abcd}^+ = C_{\Omega,\Omega}^{ab;cd} + \frac{1}{4} O_{ab}^{gh} V_{ghij} O_{cd}^{ij} + \dot{Q}_{cd} O_{ab}^{gh} D_{ghcd} \\ + \dot{Q}_{ab} O_{cd}^{gh} D_{ghab} + 4\dot{Q}_{ab} \dot{Q}_{cd} A_{abcd}, \quad (25)$$

$$B_{abcd}^+ = 4\mathbb{E}[I_a I_b I_c I_d] H_{ac} H_{bd} + 4\dot{Q}_{ab} \dot{Q}_{cd} B_{abcd}. \quad (26)$$

These equations are deterministic and finite-depth exact for the leading width coefficient. Their only source integrals are positive-orthant Gaussian moments of degree at most four and dimension at most four. Our implementation uses Tallis integration by parts, exact arcsine orthant probabilities through dimension three, and the one-dimensional Plackett path formula in dimension four. Rank-one and rank-two singular covariances are evaluated by exact half-normal and angular reductions. The largest reported raw quadrature error through depth 2000 is below 8×10^{-10} .

5 Exact componentwise depth powers

Let

$$s_a = \sqrt{K_{aa}}, \quad \delta_{ab} = \mathbf{1}_{a \neq b}, \quad h_{ab} = \frac{s_a s_b}{2(1 + 3\delta_{ab})}. \quad (27)$$

The diagonal critical variance is constant in depth. Distinct correlations coalesce, but slowly.

Lemma 5.1 (Critical correlation and NTK drift). *For fixed distinct non-collinear inputs,*

$$\theta_{ab,L} = \frac{3\pi}{L + (3/2) \log L + C_{ab} + o(1)}, \quad (28)$$

$$H_{aa,L} = \frac{s_a^2}{2} L + O(1), \quad (29)$$

$$H_{ab,L} = \frac{s_a s_b}{8} \left[L + \frac{3}{2} \log L \right] + O(1), \quad a \neq b. \quad (30)$$

The logarithm in (28) is already large enough at $L \leq 30$ to bias log-log regressions. More importantly, the collision Hessian in O forces one to retain transverse subleading modes. Keeping only the common rank-one mode $V \sim 5L$ gives wrong coefficients for D and A .

Theorem 5.2 (Attained componentwise powers and coefficients). *Fix a, b, c, d with positive s_a, s_b, s_c, s_d , and assume every pair of distinct input labels appearing in the component has initial correlation strictly between -1 and 1 . Then the critical ReLU recursions (22)–(26) satisfy*

$$V_{abcd,L} = v_{abcd} L + o(L), \quad D_{abcd,L} = d_{abcd} L^2 + o(L^2), \\ F_{abcd,L} = f_{abcd} L^2 + o(L^2), \quad A_{abcd,L} = a_{abcd} L^3 + o(L^3), \\ B_{abcd,L} = b_{abcd} L^3 + o(L^3), \quad (31)$$

Table 1: Exact normalized leading coefficients for all plotted components.

	0000	0101	0022	0103	0023	0123
V/L	5	5	5	5	5	5
D/L^2	3/2	3/5	3/2	3/5	3/5	3/5
F/L^2	1/2	1/2	1/8	1/8	1/8	1/8
A/L^3	7/12	227/2880	7/12	149/1920	47/240	37/480
B/L^3	1/6	1/18	1/96	1/72	1/192	1/288

Table 2: Paper Monte Carlo slopes / deterministic-recursion slopes over the same depth window. The last column is the exact asymptotic power.

	0000	0101	0022	0103	0023	0123	exact
V	1.19/1.075	1.31/1.380	1.49/1.585	1.42/1.376	1.47/1.612	1.55/1.510	1
D	2.06/2.075	2.34/2.284	2.67/2.651	2.35/2.271	2.55/2.540	2.49/2.431	2
F	2.24/2.075	2.35/2.191	2.69/2.428	2.20/1.961	2.67/2.405	2.24/2.021	2
A	3.16/3.096	3.39/3.181	3.74/3.752	3.39/3.164	3.67/3.608	3.54/3.343	3
B	3.28/3.135	3.34/2.975	3.78/3.426	3.16/2.696	3.76/3.248	3.43/2.827	3

where

$$v_{abcd} = 5s_a s_b s_c s_d, \quad (32)$$

$$d_{abcd} = \frac{3s_a s_b s_c s_d}{2 + 3\delta_{cd}}, \quad (33)$$

$$f_{abcd} = s_a s_c h_{bd} = \frac{s_a s_b s_c s_d}{2(1 + 3\delta_{bd})}, \quad (34)$$

$$a_{abcd} = s_a s_b s_c s_d \times \begin{cases} 7/12, & \delta_{ab} + \delta_{cd} = 0, \\ 47/240, & \delta_{ab} + \delta_{cd} = 1, \\ 227/2880, & \delta_{ab} + \delta_{cd} = 2, \{a, b\} = \{c, d\}, \\ 149/1920, & \delta_{ab} + \delta_{cd} = 2, |\{a, b\} \cap \{c, d\}| = 1, \\ 37/480, & \delta_{ab} + \delta_{cd} = 2, \{a, b\} \cap \{c, d\} = \emptyset, \end{cases} \quad (35)$$

$$b_{abcd} = \frac{2h_{ac}h_{bd}}{3(1 + \delta_{ab} + \delta_{cd})}. \quad (36)$$

Every coefficient is strictly positive. Hence the five powers are attained for every component, including all components in the published plots.

For a single input, Theorem 5.2 is the leading part of exact polynomials. Writing $q = K_{11}$,

$$\begin{aligned} H_L &= qL/2, & V_L &= 5q^2(L-1), \\ D_L &= \frac{3}{2}q^2L(L-1), & F_L &= \frac{1}{2}q^2L(L-1), \\ A_L &= \frac{q^2}{12}L(L-1)(7L-2), & B_L &= \frac{q^2}{12}L(L-1)(2L-1). \end{aligned} \quad (37)$$

Table 1 specializes the theorem to the six plotted index strings, after division by $s_a s_b s_c s_d$.

6 What the original plot exponents measure

The original experiments sample finite networks, estimate each tensor by Monte Carlo, and regress $\log |T_L|$ on $\log L$ over a selected subset of depths up to 30. Those slopes are valuable effective exponents, but they are not analytic asymptotic powers. We apply the identical windows to the deterministic recursion (22)–(26).

The deterministic recursion closely explains V, D, A . The systematic positive offset in several finite-network F, B slopes is a finite-width or higher-order effect: it is absent from the exact leading-width coefficient and must not be reinterpreted as a new critical power. Figure 2 shows the separation visually; Figure 3 follows the exact recursion to depth 2000.

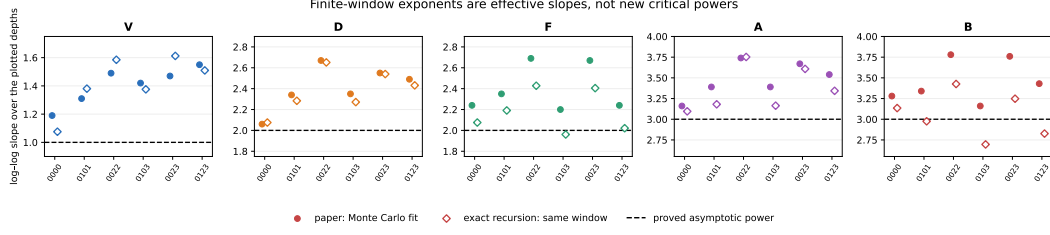


Figure 2: The published Monte Carlo regression slopes, the deterministic leading-width recursion under the same finite windows, and the proved asymptotic powers.

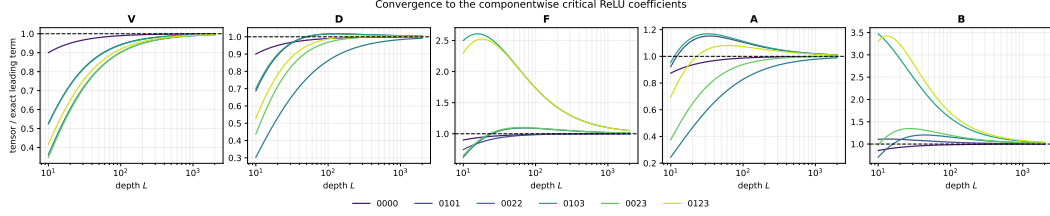


Figure 3: Each component divided by its exact leading term from Theorem 5.2. Slow, nonmonotone convergence explains why depth 30 produces noninteger effective slopes.

7 All-valence low-rank calculus

We now compute the rank vertices exactly. Let $\mathcal{P}_2(2k)$ denote the perfect matchings of $\{1, \dots, 2k\}$, and let $\tau = (1, 2)(3, 4) \dots (2k-1, 2k)$ pair the two legs belonging to each Gram entry. For a pairing π , write δ_I^π for the product of Kronecker deltas identifying external row indices along π , and $c(\pi \vee \tau)$ for the number of connected components of the graph formed by the two matchings.

7.1 The actual Gaussian MMNN vertex

Let $X \in \mathbb{R}^{n \times r}$ have iid standard-normal entries and $G = XX^\top / r$. The following theorem is both the exact moment rule and a complete connected Feynman rule.

Theorem 7.1 (Gaussian Gram Wick rule). *For arbitrary fixed k and concrete external indices $I = (i_1, j_1, \dots, i_k, j_k)$,*

$$\mathbb{E} \prod_{t=1}^k G_{i_t j_t} = \sum_{\pi \in \mathcal{P}_2(2k)} \delta_I^\pi r^{c(\pi \vee \tau) - k}, \quad (38)$$

$$\text{cum}(G_{i_1 j_1}, \dots, G_{i_k j_k}) = r^{1-k} \sum_{\substack{\pi \in \mathcal{P}_2(2k) \\ c(\pi \vee \tau) = 1}} \delta_I^\pi. \quad (39)$$

The formulas are finite- r identities, not asymptotic expansions.

In particular,

$$\mathbb{E} G_{ij} = \delta_{ij}, \quad (40)$$

$$\text{Cov}(G_{ij}, G_{kl}) = \frac{1}{r} (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk}), \quad (41)$$

$$\text{cum}_k(G_{11}) = \frac{2^{k-1} (k-1)!}{r^{k-1}}. \quad (42)$$

Thus the rank expansion is controlled at every valence, including its non-Gaussian corrections. Equation (39) is especially useful computationally: a cumulant is obtained by retaining only connected pairing diagrams.

7.2 Whitenened orientation and orthogonal Weingarten calculus

Let $U \in \mathbb{R}^{n \times r}$ be Haar on the Stiefel manifold $U^\top U = I_r$, and set $P = UU^\top$, $G = (n/r)P$. For $\pi, \sigma \in \mathcal{P}_2(2k)$, define the orthogonal Gram matrix

$$\mathcal{G}_{\pi\sigma}^{O(n)} = n^{c(\pi \vee \sigma)}, \quad \text{Wg}_n^O = (\mathcal{G}^{O(n)})^{-1}. \quad (43)$$

Theorem 7.2 (All-valence Haar-projector rule). *For arbitrary fixed k for which the orthogonal Gram matrix (43) is invertible (in particular, $n \geq k$),*

$$\mathbb{E} \prod_{t=1}^k G_{i_t j_t} = \left(\frac{n}{r}\right)^k \sum_{\pi, \sigma \in \mathcal{P}_2(2k)} \delta_I^\pi \text{Wg}_n^O(\pi, \sigma) r^{c(\sigma \vee \tau)}. \quad (44)$$

Joint cumulants follow by exact Möbius inversion on set partitions. At $r = n$, $G = I_n$ pathwise and every centered cumulant vanishes exactly.

Corollary 7.3 (Exact full-rank endpoint). *For the Stiefel-whitenened model at $r = n$, $G = I_n$ pathwise. Hence every connected rank vertex of valence at least two vanishes, and the rank-colored diagrammatics reduces to the dense Euclidean tangent calculus. Equivalently, $M = UA$ is an orthogonal reparameterization of a dense trainable matrix. For an iid Gaussian square W , the function parameterization is full rank almost surely but its Euclidean NTK is not this endpoint: the Wishart radial vertices of $WW^\top/n$ remain.*

At second order, Theorem 7.2 gives

$$\text{Cov}(G_{ij}, G_{kl}) = \gamma_{n,r} \left(\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk} - \frac{2}{n} \delta_{ij} \delta_{kl} \right), \quad (45)$$

with $\gamma_{n,r}$ from (11). For example,

$$\text{Var}(G_{11}) = \frac{2(n-r)}{r(n+2)}, \quad (46)$$

$$\text{cum}_3(G_{11}) = \frac{8(n-r)(n-2r)}{r^2(n+2)(n+4)}. \quad (47)$$

Our symbolic implementation constructs Wg_n^O by exact SymPy inversion, compresses coefficients by orthogonal coset type, and checks diagonal moments against the beta law

$$\mathbb{E} G_{11}^k = \left(\frac{n}{r}\right)^k \frac{(r/2)_k}{(n/2)_k}. \quad (48)$$

Remark 7.4 (Why both theorems are needed). *For Gaussian $X = QR$, the Stiefel orientation Q is independent of the Wishart radial factor R , and*

$$XX^\top/r = Q(RR^\top/r)Q^\top.$$

Theorem 7.2 integrates the orientation exactly; Theorem 7.1 integrates orientation and radius together. The projector-only surrogate amounts to replacing $RR^\top/r$ by a scalar identity. This is a change of tangent metric, not a harmless rewriting of the Euclidean MMNN.

8 MMNN analogues and their exact depth exponents

We now lift the definitions (12)–(14), rather than merely reusing their names. For component indices $a, b, c, d \in [r_\ell]$, define the full (not $1/n$ -stripped) MMNN tensors by

$$\text{cum}(h_1^a, h_2^b, h_3^c, h_4^d) = \delta_{ab} \delta_{cd} \mathcal{V}_{1234} + \delta_{ac} \delta_{bd} \mathcal{V}_{1324} + \delta_{ad} \delta_{bc} \mathcal{V}_{1423}, \quad (49)$$

$$\text{cum}(h_1^a, h_2^b, \Delta \Theta_{34}^{cd}) = \delta_{ab} \delta_{cd} \mathcal{D}_{1234} + \delta_{ac} \delta_{bd} \mathcal{F}_{1324} + \delta_{ad} \delta_{bc} \mathcal{F}_{1423}, \quad (50)$$

$$\text{cum}(\Delta \Theta_{12}^{ab}, \Delta \Theta_{34}^{cd}) = \delta_{ab} \delta_{cd} \mathcal{A}_{1234} + \delta_{ac} \delta_{bd} \mathcal{B}_{1324} + \delta_{ad} \delta_{bc} \mathcal{B}_{1423}. \quad (51)$$

These decompositions follow from orthogonal invariance of the Gaussian component initialization. They are the literal MMNN counterparts of the five non-MMNN tensors, with h^a replacing a dense neural channel and the matrix NTK from (5) replacing the scalar-channel NTK.

For homogeneous widths $n_\ell = n$ and ranks $r_\ell = r$, each tensor has a two-color connected expansion

$$\mathcal{T}_L = \frac{1}{n}T_L^{[n]} + \frac{1}{r}T_L^{[r]} + O_L(n^{-2} + n^{-1}r^{-1} + r^{-2}), \quad \mathcal{T} \in \{\mathcal{V}, \mathcal{D}, \mathcal{F}, \mathcal{A}, \mathcal{B}\}. \quad (52)$$

The blue coefficient $T^{[n]}$ contains one connected empirical-feature vertex; the red coefficient $T^{[r]}$ contains one connected Gram vertex. Mixed blue–red graphs start one order later. In the Stiefel-whitened model, $r^{-1}T^{[r]}$ in (52) is replaced by $\gamma_{n,r}T^{[S]}$, with the trace subtraction in (45) included in $T^{[S]}$.

The rank coefficients are computable without fitting. Let $\mathfrak{D}_{T,\ell}^{ij;kl}$ denote the two-Gram-leg Fréchet derivative obtained by differentiating the pathwise update (5) and the corresponding activation moment defining T . The one-rank-vertex source is exactly

$$S_{T,\ell}^{[r]} = \sum_{ij,kl} \mathfrak{D}_{T,\ell}^{ij;kl} (\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}), \quad (53)$$

$$S_{T,\ell}^{[S]} = \sum_{ij,kl} \mathfrak{D}_{T,\ell}^{ij;kl} \left(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk} - \frac{2}{n}\delta_{ij}\delta_{kl} \right). \quad (54)$$

Equations (53) and (54) are respectively the coefficients of $1/r$ and $\gamma_{n,r}$; all higher-valence sources follow from Theorems 7.1 and 7.2. Thus Wick or Weingarten calculus computes the amplitude tensor, including its index dependence, rather than guessing it from a scalar rank count.

At first connected order, differentiation of the pathwise recursion gives the triangular transported hierarchy, separately for $c \in \{[n], [r], [S]\}$,

$$V_{\ell+1}^c = \mathcal{L}_{V,\ell}V_\ell^c + S_{V,\ell}^c, \quad (55)$$

$$C_{\ell+1}^c = \mathcal{L}_{C,\ell}C_\ell^c + \mathcal{M}_{CV,\ell}V_\ell^c + S_{C,\ell}^c, \quad C = (D, F), \quad (56)$$

$$N_{\ell+1}^c = \mathcal{L}_{N,\ell}N_\ell^c + \mathcal{M}_{NC,\ell}C_\ell^c + \mathcal{M}_{NV,\ell}V_\ell^c + S_{N,\ell}^c, \quad N = (A, B). \quad (57)$$

All operators here are explicit Gaussian activation expectations. In the dense-compatible bias-free normalization, the blue specialization is exactly (22)–(26). Retaining the trainable component bias c_ℓ adds a local identity source but does not alter the transport powers below.

Theorem 8.1 (MMNN two-color depth powers). *Consider critical ReLU MMNNs and a fixed finite, non-collinear input set. Let $n = n(L)$, $r = r(L)$ tend to infinity with $n/r \rightarrow \rho \in (0, \infty)$, and put $\epsilon_{n,r} = n^{-1} + r^{-1}$. Uniformly in the perturbative depth window $L^3\epsilon_{n,r} \rightarrow 0$, the tensors (49)–(51) obey*

$$\mathcal{V}_L = L \left(\frac{v^{[n]}}{n} + \frac{v^{[r]}}{r} + o(\epsilon_{n,r}) \right), \quad (58)$$

$$(\mathcal{D}_L, \mathcal{F}_L) = L^2 \left(\frac{(d^{[n]}, f^{[n]})}{n} + \frac{(d^{[r]}, f^{[r]})}{r} + o(\epsilon_{n,r}) \right), \quad (59)$$

$$(\mathcal{A}_L, \mathcal{B}_L) = L^3 \left(\frac{(a^{[n]}, b^{[n]})}{n} + \frac{(a^{[r]}, b^{[r]})}{r} + o(\epsilon_{n,r}) \right). \quad (60)$$

The same statement holds for the whitened Stiefel model after replacing $r^{-1}t^{[r]}$ by $\gamma_{n,r}t^{[S]}$. The blue coefficients are strictly nonzero componentwise and, in the dense-compatible normalization, equal the coefficients in Theorem 5.2; hence all five displayed powers are attained. In a pure-rank limit $n/r \rightarrow \infty$, the stated powers remain upper bounds and are attained precisely in sectors where the corresponding projected source (53) is nonzero.

Proof. At first connected order a diagram contains exactly one blue or one red connected vertex; this proves the additive decomposition (52). Conditioning on the preceding layer and applying the law of total cumulance gives (55)–(57). The conditional blue cumulant is the ordinary empirical-average cumulant, whereas the red conditional cumulant is (41); this proves (53). Equation (54) follows identically from (45).

By Lemma 5.1, the critical gate transport has the form $I + \mathcal{K}/L + O(\log L/L^2)$. The local source in the V block is $O(1)$; the mixed block has one possible NTK path and source $O(L)$; the NTK-covariance

Table 3: Exact critical depth exponents. The dense column gives the physical connected cumulant, not the $1/n$ -stripped coefficient. Constants and sample-index contractions are suppressed.

tensor sector	exponent	dense/non-MMNN	Gaussian MMNN	Stiefel MMNN
V	1	L/n	$L(1/n + 1/r)$	$L(1/n + \gamma_{n,r})$
D, F	2	L^2/n	$L^2(1/n + 1/r)$	$L^2(1/n + \gamma_{n,r})$
A, B	3	L^3/n	$L^3(1/n + 1/r)$	$L^3(1/n + \gamma_{n,r})$

block has two such paths and source $O(L^2)$. The collision generators are finite dimensional and have eigenvalues $0, -3, -6$. Applying the discrete transport limit (145) with $p = 1, 2, 3$ yields respectively L, L^2, L^3 . More explicitly, for each color and block,

$$t_T^c = (p_T I - \mathcal{K}_T)^{-1} s_T^c, \quad (p_V, p_D, p_F, p_A, p_B) = (1, 2, 2, 3, 3), \quad (61)$$

where s_T^c is the collision limit of the exact source above. This is an exact finite-dimensional computation of every leading amplitude.

A diagram with at least two connected vertices is smaller by another factor $\epsilon_{n,r}$; its worst secular growth relative to first order is $O(L^3 \epsilon_{n,r})$. The stated depth-window condition therefore makes their sum negligible. Finally, the blue dense-compatible source reduces to the non-MMNN recursion and its coefficients are strictly positive by Theorem 5.2, proving attainment. If the blue sector is removed asymptotically, the same argument proves the claimed source-projection criterion for the red sector. $\square$

Table 3 is the requested separation. Equal depth powers do not mean equal theories: MMNNs have an additional rank amplitude, mixed higher-order diagrams, and a different breakdown scale. At $r = n$, the Stiefel rank amplitude vanishes exactly and the dense tangent theory is recovered. At $r = n$ with an iid Gaussian frozen factor, the Wishart radial amplitude remains; full matrix rank alone therefore does not recover the dense Euclidean NTK.

9 Benigni–Paquette blocks and multilayer concatenation

The preceding Feynman calculation controls finite-width and finite-rank deviations at fixed sample size. The quadratic sample–parameter regime of Benigni and Paquette [6] supplies a complementary, nonperturbative bulk law. We now connect the two without identifying a two-layer network with an entire deep network.

For a unit output direction $u \in \mathbb{R}^{r_L}$, let

$$P_{\ell \rightarrow L}(x) = J_L(x) J_{L-1}(x) \cdots J_{\ell+1}(x), \quad P_{L \rightarrow L} = I. \quad (62)$$

Define the tangent feature born at layer ℓ by

$$\omega_{\ell,u}(x) = \begin{bmatrix} n_\ell^{-1/2} \phi(z_\ell(x)) \otimes P_{\ell \rightarrow L}(x)^\top u \\ P_{\ell \rightarrow L}(x)^\top u \end{bmatrix} \in \mathbb{R}^{r_\ell(n_\ell+1)}, \quad \omega_{L,u}(x) = \bigoplus_{\ell=1}^L \omega_{\ell,u}(x). \quad (63)$$

The second entry records the trainable component bias and is omitted in the bias-free model.

Proposition 9.1 (Exact concatenated Gram representation). *For every finite MMNN realization and all inputs x, y ,*

$$u^\top \Theta_L(x, y) u = \langle \omega_{L,u}(x), \omega_{L,u}(y) \rangle. \quad (64)$$

Consequently the scalar-output sample NTK on N inputs is the Gram matrix of a concatenation of L low-rank tangent blocks, with total tangent dimension

$$d_L = \sum_{\ell=1}^L r_\ell(n_\ell + 1). \quad (65)$$

Proof. The derivative with respect to one entry of the trainable right factor is

$$\frac{\partial u^\top h_L(x)}{\partial (A_\ell)_{a\mu}} = \frac{\phi(z_{\ell,\mu}(x))}{\sqrt{n_\ell}} u^\top P_{\ell \rightarrow L}(x) e_a,$$

whereas differentiation with respect to c_ℓ^a gives $u^\top P_{\ell \rightarrow L}(x) e_a$. Taking the Euclidean inner product over (a, μ) , then summing over ℓ , proves (64). This is also the iterated form of the pathwise recursion (5). $\square$

For one two-layer block, the tangent row of Benigni and Paquette [6] is

$$\omega_{\text{BP}}(x) = x \otimes D\phi(x^\top W / \sqrt{d}) \in \mathbb{R}^{dp}; \quad (66)$$

its normalized Gram matrix is exactly

$$\frac{XX^\top}{d} \odot \frac{\phi(XW/\sqrt{d})D^2\phi(XW/\sqrt{d})^\top}{p}.$$

Equation (63) has the same Kronecker-Gram form, with $P_{\ell \rightarrow L}^\top u$ playing the transported-gain role of the weighted derivative features. The identification is exact algebraically; the probabilistic assumptions of the two-layer theorem must still be checked for the transported features.

Let $x_1, \dots, x_N$ be iid data and condition on all network matrices. The rows $\omega_{L,u}(x_i)$ are then iid. Center them, put

$$\xi_L(x) = \sqrt{d_L} [\omega_{L,u}(x) - \mathbb{E}_x \omega_{L,u}(x)], \quad \mathcal{Q}_L = \mathbb{E}_x [\xi_L(x) \xi_L(x)^\top \mid \{W, A\}],$$

and let π_L be the limiting empirical eigenvalue distribution of the block population covariance $\mathcal{Q}_L$.

Theorem 9.2 (Conditional multilayer Marchenko–Pastur law). *Suppose $N/d_L \rightarrow \gamma \in (0, \infty)$, the empirical law of $\mathcal{Q}_L$ converges in probability to a deterministic compactly supported law π_L , its operator norm is tight, and the centered tangent rows satisfy the Bai–Zhou quadratic-form concentration condition. Then the empirical spectral distribution of the centered MMNN sample NTK converges in probability to*

$$\mu_L = \mu_{\text{MP}}^\gamma \boxtimes \pi_L. \quad (67)$$

The discarded conditional mean changes the sample Gram matrix by rank at most two and therefore does not change its limiting bulk law.

Proof. By Proposition 9.1, the centered sample NTK is $d_L^{-1} \Xi_L \Xi_L^\top$, where the rows of Ξ_L are conditionally iid with covariance $\mathcal{Q}_L$. The Bai–Zhou sample-covariance theorem, in the form used by Benigni and Paquette [6], gives (67). Centering a Gram matrix changes it by the sum of two rank-one cross terms and one rank-one mean term, whose total rank is at most two. $\square$

The new multilayer content is therefore the population covariance $\mathcal{Q}_L$, not a second application of the MP map. It is a block matrix

$$\mathcal{Q}_L = (\mathcal{Q}_{\ell k})_{\ell, k \leq L}, \quad \mathcal{Q}_{\ell k} = d_L \text{Cov}_x(\omega_{\ell, u}(x), \omega_{k, u}(x) \mid \{W, A\}). \quad (68)$$

The diagonal blocks are transported Benigni–Paquette covariance tensors. The off-diagonal blocks record correlations between tangent features born at different depths and are computed by the same colored Feynman rules as (55)–(57).

If the off-diagonal blocks are negligible in normalized Hilbert–Schmidt norm, write $d_\ell = r_\ell(n_\ell + 1)$ and $\alpha_\ell = d_\ell/d_L$. If the naturally normalized diagonal block has limiting law π_ℓ^{BP} , then

$$\pi_L = \sum_{\ell=1}^L \alpha_\ell (\alpha_\ell^{-1} \cdot)_\# \pi_\ell^{\text{BP}}, \quad \mu_L = \mu_{\text{MP}}^\gamma \boxtimes \pi_L. \quad (69)$$

When cross-layer blocks persist, their full block law must be retained; replacing it by a free additive convolution is an additional asymptotic freeness hypothesis, not a consequence of concatenation alone.

Corollary 9.3 (Closed homogeneous concatenation law). *Assume the hypotheses of Theorem 9.2, negligible off-diagonal population blocks, and L statistically identical tangent blocks of dimension dp . If $N/(dp) \rightarrow \gamma_1$ and the one-block population covariance law is $\mu_{\nu, \varphi}$ in the notation of Benigni and Paquette, then*

$$\mu_{\Theta_L} = \mu_{\text{MP}}^{\gamma_1/L} \boxtimes (L \cdot)_\# \mu_{\nu, \varphi}. \quad (70)$$

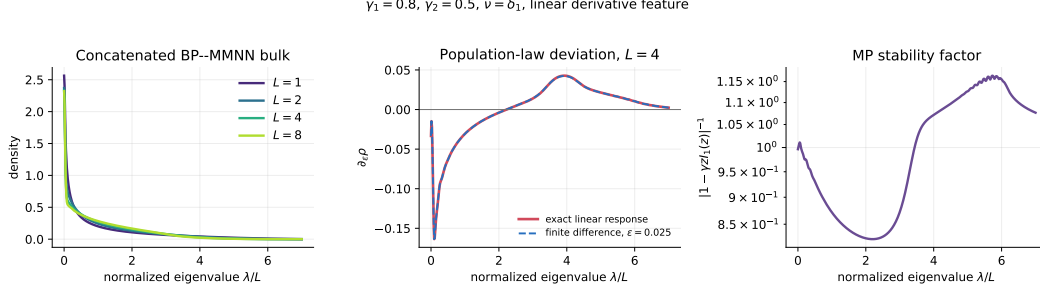


Figure 4: Conditional exact quadratic-scaling spectrum for homogeneous concatenated Benigni–Paquette tangent blocks. Left: the normalized spectrum from (70). Center: the analytic population-law response (75) and a finite difference. Right: the MP stability factor that filters a Feynman correction. We use $\gamma_1 = 0.8$, $\gamma_2 = 0.5$, $\nu = \delta_1$, and the explicit linear derivative-feature law.

More generally, for transported squared gains τ_ℓ^2 and possibly different one-block laws,

$$\mu_{\Theta_L} = \mu_{\text{MP}}^{\gamma_1/L} \boxtimes \left[\frac{1}{L} \sum_{\ell=1}^L (L\tau_\ell^2 \cdot)_{\#} \mu_{\nu_\ell, \varphi_\ell} \right]. \quad (71)$$

In the explicit Benigni–Paquette case $\nu = \delta_1$ or $\varphi(x) = ax + b$,

$$\mu_{\nu, \varphi} = \text{Law}(\alpha_\varphi^2 \chi + \beta_\varphi^2 \psi), \quad (72)$$

where

$$\begin{aligned} \text{Law}(\chi) &= \frac{\gamma_2}{2} (\mu_{\text{MP}}^{\gamma_2} \boxtimes \nu) * (\mu_{\text{MP}}^{\gamma_2} \boxtimes \nu) \\ &\quad + (1 - \gamma_2) (\mu_{\text{MP}}^{\gamma_2} \boxtimes \nu) + \frac{\gamma_2}{2} \delta_0. \end{aligned} \quad (73)$$

Equations (70) and (73) therefore compute the concatenated Gram–NTK bulk using only scalar measure operations and the MP fixed point.

Proof. Here $d_L = Ldp$, hence $N/d_L \rightarrow \gamma_1/L$ and $\alpha_\ell = 1/L$. Equation (69) gives $\pi_L = (L \cdot)_{\#} \mu_{\nu, \varphi}$ in the homogeneous unit-gain case and the mixture in (71) otherwise. Theorem 9.2 supplies the outer MP map. The last statement is the explicit population law of Benigni and Paquette. $\square$

9.1 Spectral deviations around the quadratic-scaling bulk

The Benigni–Paquette law can be used as the base spectrum rather than replaced by a Wigner base. Let $s(z)$ be the Stieltjes transform of $\mu_{\text{MP}}^{\gamma} \boxtimes \pi$, so that

$$s(z) = \int \frac{d\pi(t)}{t[1 - \gamma(1 + zs(z))] - z}. \quad (74)$$

Put $a(z) = 1 - \gamma(1 + zs(z))$ and $D(t, z) = ta(z) - z$. For a signed population-law perturbation $\delta\pi$ of total mass zero, differentiation of (74) gives the exact linear response

$$\delta s(z) = \frac{\int D(t, z)^{-1} d(\delta\pi)(t)}{1 - \gamma z \int t D(t, z)^{-2} d\pi(t)}. \quad (75)$$

Thus width- and rank-colored diagrams first perturb the block covariance law π_L ; equation (75) propagates that perturbation to the bulk density $\delta\rho(\lambda) = \pi^{-1} \Im \delta s(\lambda + i0)$.

Sample-to-sample spectral fluctuations retain the full tensor information. If $R_z = (K_{\text{BP}} - zI)^{-1}$, the first resolvent variation is

$$\delta m_N(z) = -\frac{1}{N} \text{Tr}(R_z \Delta K R_z), \quad (76)$$

and its covariance follows by contracting two copies of R_z^2 with the A/B tensor, including the additional $1/r$ Gram or Weingarten vertex. Consequently its perturbative scale is $L^3(1/n + 1/r)$, while its spectral shape is filtered by the stability denominator in (75). Near a quadratic-scaling spectral edge this denominator may be small, amplifying a diagrammatically small finite-rank correction.

Remark 9.4 (ReLU regularity). *The theorem of Benigni and Paquette [6] assumes that the derivative feature $\phi = \sigma'$ is pseudo-Lipschitz. The exact ReLU gate $\mathbf{1}_{x>0}$ is discontinuous and is not covered verbatim. Equations (64) and (68) remain exact for ReLU, but the quadratic-scaling limit then requires either a smoothed gate followed by a uniform limit or a separate universality proof.*

10 Neural quadratic forms as an exact Landau resummation

We now integrate the neural quadratic form (NQF) of Liu Ziyin et al. [8] with the preceding finite-width and finite-rank diagrams. To prevent a serious notation collision, throughout this section $\mathcal{A}_x \in \mathbb{R}^{p \times p}$ denotes the NQF *structure matrix*, $M = WW^\top$ denotes its dynamic order parameter, and $\mathcal{A}_{abcd}, \mathcal{B}_{abcd}$ continue to denote the four-sample NTK-covariance tensors of Section 8. The frozen MMNN metric $G = XX^\top/r$ is yet another Gram matrix. Algebraically, G and M have the same double-line indices; dynamically they have different roles: G is quenched, while $M(t)$ is trained.

10.1 Symmetry, the normal form, and the MMNN dictionary

Let $W = (w_1, \dots, w_d) \in \mathbb{R}^{p \times d}$ collect exchangeable components. The zero-gradient-at-zero (ZGZ) condition of Liu Ziyin et al. [8] is

$$\nabla_{w_i} f_x(w_1, \dots, w_d)|_{w_i=0} = 0 \quad \text{for every value of the other components.} \quad (77)$$

Proposition 10.1 (Symmetry-derived local vertex). *If f_x is C^3 , invariant under S_d , and satisfies (77), then, locally at the symmetric origin,*

$$f_x(W) = f_x(0) + \text{Tr}(M\mathcal{A}_x) + R_{3,x}(W), \quad M = WW^\top, \quad |R_{3,x}(W)| \leq C_x \|W\|^3. \quad (78)$$

If each component also has an independent sign symmetry $w_i \mapsto -w_i$, the remainder is $O(\|W\|^4)$. Without ZGZ the complete quadratic normal form is

$$f_x(W) = f_x(0) + g_x^\top \mu + \mu^\top \mathcal{B}_x \mu + \text{Tr}(M\mathcal{A}_x) + O(\|W\|^3), \quad \mu = W\mathbf{1}. \quad (79)$$

Proof. Write the Taylor Hessian in $p \times p$ blocks H_{ij} . Permutation symmetry makes all first derivatives equal, all diagonal blocks equal, and all off-diagonal blocks equal. Because the left-hand side of (77) is identically zero as a function of the other components, the common first derivative and every mixed block H_{ij} , $i \neq j$, vanish. Setting $\mathcal{A}_x = H_{ii}/2$ leaves $\frac{1}{2} \sum_i w_i^\top H_{ii} w_i = \text{Tr}(WW^\top \mathcal{A}_x)$. Taylor's theorem with locally bounded third derivative gives the stated remainder. Independent sign symmetry eliminates every monomial containing an odd power of a component, hence all cubic terms. If ZGZ is removed, the common first derivative gives $g_x^\top \mu$, while the common mixed Hessian can be rewritten as a quadratic form in μ plus a redefinition of the self-interaction matrix, which gives (79). $\square$

The result is a Landau construction: S_d fixes the allowed invariant, and the architecture fixes the coupling $\mathcal{A}_x$. It is essential not to infer ZGZ merely from low rank. Indeed, the one-block MMNN map $A_1 \mapsto A_1 \phi(W_1 x)$ is exactly linear in its trainable right factor and therefore occupies the $g_x^\top \mu$ sector of (79). The pure quadratic sector first appears when two trainable factors interact.

For example, consider two bias-free MMNN blocks, a scalar readout u , a smooth activation with $\phi(0) = 0$, and let $s_1(x) = \phi(W_1 x / \sqrt{r_0})$. At $(A_1, A_2) = (0, 0)$,

$$u^\top h_2(x) = \frac{\phi'(0)}{\sqrt{n_1 n_2 r_1}} u^\top A_2 W_2 A_1 s_1(x) + O(\|(A_1, A_2)\|^3). \quad (80)$$

After vectorizing A_1, A_2 , the displayed bilinear form is $b^\top \mathcal{C}_x a$, hence it has structure matrix

$$\mathcal{A}_x = \frac{1}{2} \begin{bmatrix} 0 & \mathcal{C}_x^\top \\ \mathcal{C}_x & 0 \end{bmatrix}. \quad (81)$$

The frozen matrix W_2 is part of $\mathcal{C}_x$. Contracting two A_1 -legs of $\mathcal{C}_x$ produces precisely $G = W_2 W_2^\top / r_1$; thus Theorems 7.1 and 7.2 compute the radial and orientation statistics of the random Landau vertex. For three or more simultaneously vanishing right factors, the global leading degree is generally the number of blocks, not two. One must then either expand around a nonzero reference state or use the layerwise deep NQF; a global quadratic claim would be false.

10.2 Closed dynamics and the resummed dynamic NTK

We first treat the pure ZGZ model as an exact model,

$$f_\mu(W) = \text{Tr}(\mathbf{M} \mathcal{A}_\mu), \quad \mathcal{L} = \frac{1}{m} \sum_{\mu=1}^m (f_\mu - y_\mu)^2, \quad (82)$$

with symmetric $\mathcal{A}_\mu$. Define the structure-covariance operator and target matrix

$$\mathcal{K}(X) = \frac{4}{m} \sum_{\mu=1}^m \langle \mathcal{A}_\mu, X \rangle_F \mathcal{A}_\mu, \quad Y = \frac{4}{m} \sum_{\mu=1}^m y_\mu \mathcal{A}_\mu. \quad (83)$$

Theorem 10.2 (Exact quadratic-vertex resummation). *For gradient flow in (82),*

$$\dot{W} = -H(\mathbf{M})W, \quad H(\mathbf{M}) = \mathcal{K}(\mathbf{M}) - Y, \quad (84)$$

$$\dot{\mathbf{M}} = Y\mathbf{M} + \mathbf{M}Y - \mathcal{K}(\mathbf{M})\mathbf{M} - \mathbf{M}\mathcal{K}(\mathbf{M}). \quad (85)$$

The empirical NTK is, at every finite p, d, m ,

$$\Theta_{\mu\nu}(t) = 4 \text{Tr}(\mathcal{A}_\mu \mathcal{A}_\nu \mathbf{M}(t)). \quad (86)$$

Equations (85)–(86) resum every chain made only of quadratic structure vertices. No width or lazy-training limit is used.

Proof. Since $\nabla_W f_\mu = 2\mathcal{A}_\mu W$, direct differentiation of (82) gives $\dot{W} = -(4/m) \sum_\mu (f_\mu - y_\mu) \mathcal{A}_\mu W$, which is (84). Applying the product rule to $\mathbf{M} = WW^\top$ proves (85). Finally,

$$\langle \nabla_W f_\mu, \nabla_W f_\nu \rangle_F = 4 \text{Tr}(W^\top \mathcal{A}_\mu \mathcal{A}_\nu W) = 4 \text{Tr}(\mathcal{A}_\mu \mathcal{A}_\nu \mathbf{M}),$$

which proves (86). The right side of the flow depends on W only through $\mathbf{M}$, so adding another two-leg insertion cannot generate a new order parameter; induction over the number of insertions proves the resummation statement. $\square$

For general data the propagator $U(t) = \mathcal{T} \exp[-\int_0^t H(\mathbf{M}(s)) ds]$ gives $\mathbf{M}(t) = U(t)\mathbf{M}(0)U(t)^\top$, but this is a self-consistent time-ordered representation, not a scalar closed form. A genuinely closed noncommutative solution follows under the isotropic four-tensor identity

$$\frac{1}{m} \sum_{\mu} (\mathcal{A}_\mu)_{ij} (\mathcal{A}_\mu)_{kl} = \frac{c}{2} (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk}). \quad (87)$$

Then $\mathcal{K}(\mathbf{M}) = 4c\mathbf{M}$, so

$$\dot{\mathbf{M}} = Y\mathbf{M} + \mathbf{M}Y - 8c\mathbf{M}^2, \quad \mathbf{M}(t) = e^{Yt} \mathbf{M}_0 [I + 8c\Phi(t)\mathbf{M}_0]^{-1} e^{Yt}, \quad \Phi(t) = \int_0^t e^{2Ys} ds. \quad (88)$$

Substitution verifies the Riccati equation. Diagrammatically, (88) is an exact matrix rainbow sum, whereas the deformed-Wigner Dyson equation of Section 11 resums a different object: ensemble fluctuations around a base kernel.

For completeness, the non-ZGZ quadratic model (79) also closes exactly, now on $(\mu, \mathbf{M})$. With $u_x = g_x + 2\mathcal{B}_x \mu$, its NTK is

$$\Theta(x, x') = d u_x^\top u_{x'} + 2u_x^\top \mathcal{A}_{x'} \mu + 2\mu^\top \mathcal{A}_x u_{x'} + 4 \text{Tr}(\mathcal{A}_x \mathcal{A}_{x'} \mathbf{M}). \quad (89)$$

This formula is the correct one-block MMNN entry point because its linear sector need not vanish.

The diagrammatic simplification can also be stated at all valences. For a matrix C , write $C_s = (C + C^\top)/2$ and $Q_C = \text{Tr}(CM) = \text{Tr}(C_s \mathbf{M})$.

Proposition 10.3 (Exact NQF initialization vertices). *Let $W_0 = \sqrt{\epsilon/d} Z$, where the entries of $Z \in \mathbb{R}^{p \times d}$ are iid standard Gaussian. For symmetric test matrices $C_1, \dots, C_q$ and $q \geq 2$,*

$$\text{cum}(Q_{C_1}, \dots, Q_{C_q}) = \frac{2^{q-1} \epsilon^q}{d^{q-1}} \sum_{\pi \in S_{q-1}} \text{Tr}(C_1 C_{\pi(2)} \cdots C_{\pi(q)}). \quad (90)$$

Here S_{q-1} permutes $\{2, \dots, q\}$. In particular,

$$\text{Cov}(f_\mu, f_\nu) = \frac{2\epsilon^2}{d} \text{Tr}(\mathcal{A}_\mu \mathcal{A}_\nu), \quad (91)$$

$$\text{Cov}(\Theta_{\mu\nu}, \Theta_{\rho\sigma}) = \frac{32\epsilon^2}{d} \text{Tr}(C_{\mu\nu} C_{\rho\sigma}), \quad C_{\mu\nu} = \frac{1}{2}(\mathcal{A}_\mu \mathcal{A}_\nu + \mathcal{A}_\nu \mathcal{A}_\mu). \quad (92)$$

Every joint V, D, F, A, B -type cumulant of NQF outputs and NTK entries is obtained from (90) by using $C = \mathcal{A}_\mu$ for an output and $C = 4C_{\mu\nu}$ for an NTK entry. For $M_0 = \epsilon(p/d)UU^\top$, the corresponding finite- (p, d) formula is exactly Theorem 7.2 followed by Möbius inversion; all connected vertices vanish pathwise at $d = p$.

Proof. Write $Q_C = (\epsilon/d) \sum_{a=1}^d z_a^\top C z_a$ with independent $z_a \sim N(0, I_p)$. The logarithmic moment-generating function of Gaussian quadratic forms is

$$-\frac{1}{2} \log \det \left(I - 2 \sum_{j=1}^q t_j C_j \right).$$

Expanding $-\log \det(I - X) = \sum_{k \geq 1} \text{Tr}(X^k)/k$ and extracting $t_1 \cdots t_q$ gives $2^{q-1} \sum_{\pi \in S_{q-1}} \text{Tr}(C_1 C_{\pi(2)} \cdots C_{\pi(q)})$ for one column. Cumulants add over independent columns and scale homogeneously, proving (90). Equations (91) and (92) follow from (86). The Stiefel claim is the projector moment rule and the definition of a connected cumulant. $\square$

Thus a full nonlinear NTK has arbitrarily high derivative vertices and a large tensor hierarchy, while a pure NQF has only a two-leg structure vertex: all higher connected initialization vertices are closed trace words of that same vertex, and all training insertions close on M . Low rank does not change the allowed words; it changes their exact coefficient from the dense limit to d^{1-q} (Gaussian) or a finite Weingarten weight (Stiefel). This is the precise sense in which the low-rank diagrams simplify.

10.3 Exact spectrum, mode ignition, and finite-rank clock disorder

Suppose the structure matrices commute, $\mathcal{A}_\mu = P \text{diag}(\lambda_{\mu 1}, \dots, \lambda_{\mu p}) P^\top$, and write $z_k = (P^\top M P)_{kk}$. Let $\Lambda = (\lambda_{\mu k})_{\mu, k}$.

Theorem 10.4 (Complete dynamic NTK spectrum in the orthogonal sector). *For commuting structure matrices,*

$$\Theta(t) = 4\Lambda \text{diag}(z_1(t), \dots, z_p(t)) \Lambda^\top. \quad (93)$$

If the columns $\lambda_k = \Lambda_{:,k}$ are mutually orthogonal, define

$$r_k = \frac{8}{m} \sum_{\mu} y_{\mu} \lambda_{\mu k}, \quad C_k = \frac{8}{m} \sum_{\mu} \lambda_{\mu k}^2. \quad (94)$$

Then

$$z_k(t) = \frac{r_k z_k(0)}{C_k z_k(0) + [r_k - C_k z_k(0)] e^{-r_k t}}, \quad (95)$$

with the continuous $r_k = 0$ limit, and the complete nonzero NTK eigenpairs are

$$v_k = \lambda_k / \|\lambda_k\|, \quad \kappa_k(t) = 4\|\lambda_k\|^2 z_k(t). \quad (96)$$

Proof. In the common basis, $\text{Tr}(\mathcal{A}_\mu \mathcal{A}_\nu M) = \sum_k \lambda_{\mu k} \lambda_{\nu k} z_k$, which proves (93); the off-diagonal entries of $P^\top M P$ drop out. Taking the diagonal of (85) gives $\dot{z}_k = (r_k - C_k z_k) z_k$ when the columns are orthogonal. Separation of variables proves (95). Finally, (93) is an orthogonal sum of rank-one matrices, which proves (96) including multiplicities and the null space. $\square$

Thus “mode ignition” is literally the birth of an NTK eigenvalue in this sector. The half-plateau time for $r_k > 0$ and $0 < z_k(0) < r_k/(2C_k)$ is exactly

$$t_{k,1/2} = \frac{1}{r_k} \log \frac{r_k - C_k z_k(0)}{C_k z_k(0)}. \quad (97)$$

The next theorem connects this dynamic spectrum to the low-rank Wick and Weingarten vertices.

Theorem 10.5 (Wishart–Weingarten ignition clocks). *Let $M_0 = W_0 W_0^\top$ and fix a structure eigenbasis. For Gaussian $W_0 \in \mathbb{R}^{p \times d}$ with iid $N(0, \epsilon/d)$ entries,*

$$z_k(0) \stackrel{d}{=} \epsilon \frac{\chi_d^2}{d}. \quad (98)$$

For the whitened Haar–Stiefel initialization $M_0 = \epsilon(p/d)UU^\top$, $U^\top U = I_d$,

$$z_k(0) \stackrel{d}{=} \epsilon \frac{p}{d} \text{Beta}\left(\frac{d}{2}, \frac{p-d}{2}\right), \quad d < p, \quad (99)$$

while $z_k(0) = \epsilon$ pathwise at $d = p$. Consequently, as $\epsilon \downarrow 0$, the Gaussian half-clock has

$$\mathbb{E} t_{k,1/2} = \frac{1}{r_k} \left[\log \frac{r_k}{C_k \epsilon} - \psi(d/2) - \log 2 + \log d \right] + O(\epsilon), \quad (100)$$

$$\text{Var}(t_{k,1/2}) = \frac{\psi_1(d/2)}{r_k^2} + O(\epsilon), \quad (101)$$

whereas the Stiefel half-clock has

$$\mathbb{E} t_{k,1/2} = \frac{1}{r_k} \left[\log \frac{r_k}{C_k \epsilon} - \log \frac{p}{d} - \psi(d/2) + \psi(p/2) \right] + O(\epsilon), \quad (102)$$

$$\text{Var}(t_{k,1/2}) = \frac{\psi_1(d/2) - \psi_1(p/2)}{r_k^2} + O(\epsilon). \quad (103)$$

In particular the Stiefel clock variance vanishes exactly at full rank, whereas the square iid-Gaussian clock retains its radial Wishart disorder.

Proof. Equation (98) is the diagonal Wishart law. A diagonal entry of a rank- d Haar projector is $\text{Beta}(d/2, (p-d)/2)$, proving (99); it is also the one-point specialization of Theorem 7.2. Insert either seed law into the exact pushforward (97). For $X = \chi_d^2/d$,

$$\mathbb{E} \log X = \psi(d/2) + \log 2 - \log d, \quad \text{Var}(\log X) = \psi_1(d/2).$$

For $X = (p/d)B$ with $B \sim \text{Beta}(d/2, (p-d)/2)$,

$$\mathbb{E} \log X = \log(p/d) + \psi(d/2) - \psi(p/2), \quad \text{Var}(\log X) = \psi_1(d/2) - \psi_1(p/2).$$

Expanding the remaining term $r_k^{-1} \log(1 - C_k \epsilon X / r_k)$ proves (100)–(103). At $d = p$, $UU^\top = I$ and the conclusion is pathwise. $\square$

If $r_k \asymp k^{-\alpha_2}$ and the loss remaining after mode k scales as $k^{-(\alpha_1-1)}$, the same ignition clock gives $\mathcal{E}(t) = \Theta(t^{-(\alpha_1-1)/\alpha_2})$ after rescaling time by $\log(1/\epsilon)$, as proved by Liu Ziyin et al. [8]. This exponent is conditional on the power-law tails of the rates and target energies: the NQF derives the controlling operator $\mathcal{A}_x$, but not the power law of its spectrum. Equations (100)–(103) add a finite-rank correction to that scaling limit.

10.4 Higher vertices, depth, and the Benigni–Paquette spectrum

Inside the pure NQF all parameter derivatives above second order vanish. Thus there are no irreducible cubic or quartic Landau vertices, and the closure in Theorem 10.2 is exact. For the original network, the Taylor remainder generates new vertices. Permutation symmetry organizes an order- q derivative by an integer partition $\lambda \vdash q$: each part records how many legs land on the same exchangeable component. A cubic form therefore introduces three-leg vertices and third empirical moments; independent component sign symmetry kills them, leaving quartic vertices and fourth moments as the

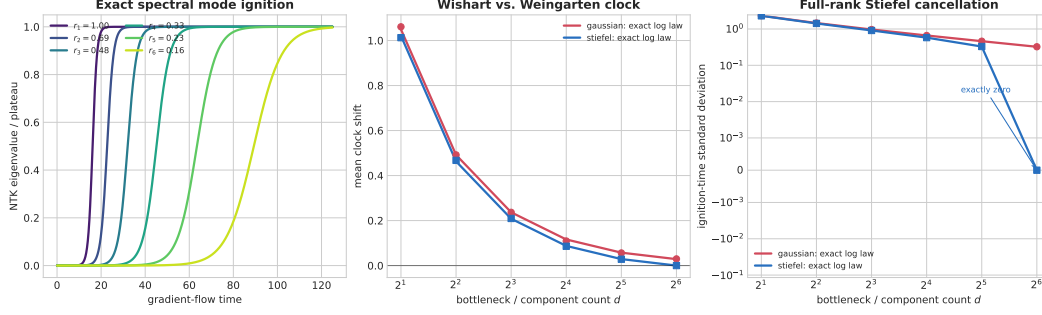


Figure 5: Exact NQF spectrum and the low-rank ignition clock. Left: all nonzero eigenvalues of the finite-sample NTK in the orthogonal-feature sector follow the logistic solution (96). Center and right: Monte Carlo points for the exact chi-square and beta seed laws versus the polygamma predictions of Theorem 10.5. The Stiefel clock disorder vanishes at $d = p = 64$; the Gaussian radial sector does not.

leading correction. Query-key attention is a particularly important example: its query/key interaction can first occur at quartic order even when its value/readout sector is quadratic. These higher vertices are not resummed by M alone and are natural sources of the additional plateaus seen outside the small-initialization regime in Liu Ziyin et al. [8].

A deep NQF defines each layer output to be quadratic in its own (μ_ℓ, M_ℓ) , with structure tensors depending on the previous layer representation. Its SGD dynamics close exactly and simultaneously on all layer pairs (μ_ℓ, M_ℓ) . This is an exact statement for the *defined deep NQF*, but only a layerwise local approximation to a generic deep MMNN. The pathwise recursion (5) remains exact for the original MMNN. Combining the two is therefore legitimate in either of two regimes:

1. the network is itself a deep NQF, in which case the layerwise order parameter closure and the tangent concatenation are both exact; or
2. every layer remains inside a controlled Taylor neighborhood, in which case the NQF supplies a local approximation to the exact pathwise concatenation, with an accumulated remainder that must be bounded.

Even in the first regime, independent logistic modes require the transported structure operators to share projectors satisfying the closure conditions of Theorem 10.4. Without that alignment the exact object is a coupled, operator-valued flow rather than a product of scalar logistics.

At a fixed training time, define the pure-NQF tangent row

$$\omega_t(x) = 2 \text{vec}(\mathcal{A}_x W(t)). \quad (104)$$

Then $\Theta_t(x, x') = \langle \omega_t(x), \omega_t(x') \rangle$ exactly. Consequently Theorem 9.2 applies at time t , conditional on its quadratic-form concentration hypotheses, with the population covariance of ω_t . This yields a time-dependent Benigni-Paquette law

$$\mu_{\Theta(t)} = \mu_{\text{MP}}^\gamma \boxtimes \pi_t, \quad (105)$$

where π_t is driven by the exact order-parameter flow. Equation (105) is conditional in general; in the empirical orthogonal-feature sector, Theorem 10.4 is stronger and already gives the finite-sample spectrum without an asymptotic MP map.

11 From tensor cumulants to eigenvalues and eigenvectors

Let $K_0 \in \mathbb{R}^{m \times m}$ be the deterministic infinite-width NTK Gram matrix on the training data, and write

$$\hat{K} = K_0 + \Delta. \quad (106)$$

The A tensor is the leading covariance of the scalar-output NTK fluctuation. In the dense model,

$$\text{Cov}(\Delta_{ab}, \Delta_{cd}) = \frac{1}{n} A_{abcd, L} + O(n^{-2}). \quad (107)$$

Table 4: Status of the NQF–MMNN integration. “Exact in NQF” never means global exactness for the original nonlinear network.

statement	status	missing hypothesis or remainder
Symmetry normal form (78)	rigorous local theorem	C^3, S_d, ZGZ ; cubic remainder
Quadratic dynamics and NTK	exact in NQF	no large-width assumption
Isotropic Riccati sum	exact in NQF	sample four-tensor isotropy
Orthogonal-mode spectrum	exact finite sample	commuting structure and orthogonal feature columns
Wishart/Stiefel clocks	exact seed laws; asymptotic moments	small- ϵ expansion only for displayed moments
One-block MMNN	exact linear normal-form sector	generally not pure ZGZ
Two-block formula (80)	local quadratic MMNN sector	smooth $\phi, \phi(0) = 0$, small right factors
Deep NQF closure	exact for defined deep NQF	local for a generic deep MMNN
Dynamic BP law (105)	conditional asymptotic theorem	population convergence and quadratic-form concentration
Power-law learning exponent	conditional asymptotic theorem	power-law rate and target spectra are assumed

In an MMNN, diagrams with one connected Gaussian Gram vertex add a term of order $1/r$, with its exact index contraction fixed by (41); higher rank cumulants are given by (39). Theorem 5.2 therefore implies the perturbative entry scale

$$\Delta_{ab} = O_{\text{rms}} \left(L^{3/2} \sqrt{\frac{c_n}{n} + \frac{c_r}{r}} \right), \quad L^3 \left(\frac{1}{n} + \frac{1}{r} \right) \ll 1, \quad (108)$$

for data-dependent contractions c_n, c_r . Beyond this regime, multiplicative finite-rank tails need not be perturbative.

11.1 General matrix Dyson equation

Suppose the leading rescaled correction is Gaussian with covariance operator

$$\mathcal{S}[R]_{ij} = \sum_{k,l} \mathbb{E}(\Delta_{ik} \Delta_{lj}) R_{kl}. \quad (109)$$

This Gaussian leading sector follows when width and rank grow so that the connected vertices of order $k \geq 3$ vanish under the spectral scaling; the exact formulas above quantify the error. The deterministic resolvent $M(z)$ is the solution with positive imaginary part of

$$-M(z)^{-1} = zI - K_0 + \mathcal{S}[M(z)]. \quad (110)$$

The limiting density is

$$\rho(\lambda) = \frac{1}{\pi m} \lim_{\eta \downarrow 0} \Im \text{Tr } M(\lambda + i\eta). \quad (111)$$

Equation (110), rather than Weingarten calculus alone, is the spectral resummation of the tensor diagrams.

11.2 Isotropic deformed-Wigner sector

If $K_0 = \text{diag}(\eta_1, \dots, \eta_m)$ and the projected covariance is isotropic GOE with off-diagonal variance σ^2/m , then M is diagonal and (110) reduces to

$$\mathbf{m}(z) = \frac{1}{m} \sum_{i=1}^m \frac{1}{\eta_i - z - \sigma^2 \mathbf{m}(z)}, \quad (112)$$

$$M_{ii}(z) = \frac{1}{\eta_i - z - \sigma^2 \mathbf{m}(z)}. \quad (113)$$

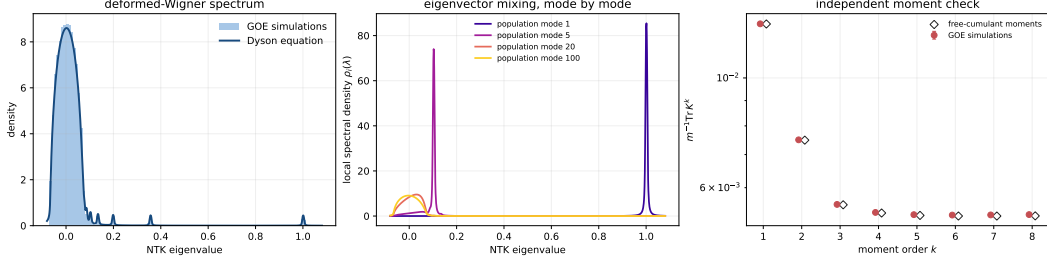


Figure 6: Left: the Dyson density versus direct GOE simulations. Middle: basis-resolved local spectral measures, which quantify eigenvector mixing. Right: free-cumulant trace moments versus independent simulations through eighth order.

The local spectral density

$$\rho_i(\lambda) = \frac{1}{\pi} \Im M_{ii}(\lambda + i0) \quad (114)$$

determines eigenvector mixing. Conditional on an eigenvalue near λ , the mean squared overlap with population mode i is

$$\mathbb{E}[|\langle e_i, u_\lambda \rangle|^2 | \lambda] = \frac{\rho_i(\lambda)}{m\rho(\lambda)}. \quad (115)$$

This satisfies the exact sum rule over i .

Trace moments are computed independently. If $r_k(\mu_0)$ are the free cumulants of the empirical signal law, then the deformed law has

$$r_k(\mu) = r_k(\mu_0) + \sigma^2 \mathbf{1}_{k=2}, \quad (116)$$

and its moments follow by summing over noncrossing partitions. Our implementation enumerates these partitions exactly. For zero signal it recovers $m_{2k} = C_k \sigma^{2k}$, the Catalan semicircle moments.

For a pure GOE block the eigenvectors are Haar orthogonal, and orthogonal Weingarten calculus gives the exact finite- m moments

$$\mathbb{E}|u_i|^{2k} = \frac{(1/2)_k}{(m/2)_k}. \quad (117)$$

Equation (117) tests a fully mixed regime; it does not assert that an arbitrary NTK correction has GOE eigenvalues.

12 Power-law data and early stopping

Kramp, Lindner, and Helias study random-feature regression with population kernel eigenvalues

$$\eta_i = \eta_1 i^{-(1+\alpha)}, \quad \alpha > 0, \quad (118)$$

and derive typical Langevin learning dynamics [7]. In the weak-regularization scaling window, the aligned bias and thermal variance are

$$\mathcal{L}_{\text{bias}}(t) = \frac{1}{2} \sum_i \eta_i e^{-2Pt\eta_i}, \quad (119)$$

$$\mathcal{L}_{\text{var}}(t) = \frac{1}{2P\beta} \sum_i (1 - e^{-2Pt\eta_i}). \quad (120)$$

Consequently

$$t_0^* = \frac{\beta}{2} \frac{\alpha}{1+\alpha}, \quad \mathcal{L}(t_0^*) \asymp (P\beta)^{-\alpha/(1+\alpha)}. \quad (121)$$

For a finite-rank NTK with eigenvalues κ_j and eigenvectors u_j , the linearized risk is instead

$$\mathcal{L}(t) = \sum_{i,j} \eta_i |\langle e_i, u_j \rangle|^2 c_t(\kappa_j), \quad c_t(\kappa) = \frac{1}{2} e^{-2Pt\kappa} + \frac{1 - e^{-2Pt\kappa}}{2P\beta\kappa}, \quad (122)$$

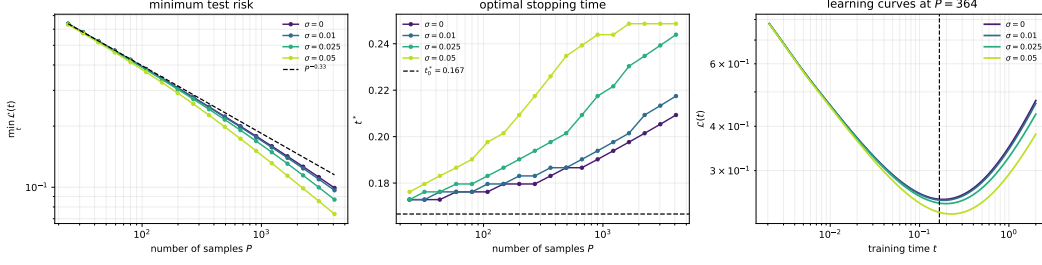


Figure 7: Dyson prediction for power-law data with $\alpha = 1/2$, $N = 4096$. The PSD correction lifts and mixes tail rates. In this ensemble the optimum moves later, demonstrating that the direction of the finite-rank early-stopping shift is not universal.

with the continuous value $c_t(0) = 1/2 + t/\beta$. The Dyson deterministic equivalent is

$$\mathcal{L}_{\text{Dyson}}(t) = \sum_i \eta_i \int c_t(\lambda) \rho_i(\lambda) d\lambda. \quad (123)$$

Because an additive Wigner approximation can become negative in the power-law tail, we evaluate a positive-semidefinite deformation

$$K_\sigma = (\text{diag}(\sqrt{\eta_i}) + \sigma W)^2, \quad W \sim \text{GOE}(m). \quad (124)$$

The eigenvectors are those of the deformed-Wigner matrix inside the square, and its eigenvalues are squared in (123). Thus the scalar Dyson equation computes the learning curve without an ad hoc Haar-tail replacement.

The stable head admits an ensemble-independent statement. The population gap satisfies

$$g_j = \eta_j - \eta_{j+1} = (1 + \alpha) \eta_1 j^{-(2+\alpha)} (1 + O(j^{-1})). \quad (125)$$

If $\|\Delta\|_{\text{op}} \leq s$, Weyl and Davis–Kahan protect modes with $g_j > 2s$, giving

$$J_{\text{mix}} \asymp \left(\frac{(1 + \alpha) \eta_1}{2s} \right)^{1/(2+\alpha)}. \quad (126)$$

At the aligned optimum the learned cutoff is $J_0 \asymp (P\beta\eta_1)^{1/(1+\alpha)}$. Requiring $J_0 \lesssim J_{\text{mix}}$ and using the Wigner-type operator scale $s \asymp L^{3/2} \sqrt{m/r}$ yields the sufficient protection law

$$r \gtrsim mL^3 (P\beta\eta_1)^{2(2+\alpha)/(1+\alpha)}, \quad (127)$$

up to data contractions and logarithmic concentration factors. Equation (127) protects the aligned law; when it fails, neither the amount nor the direction of the optimal-time shift is universal.

13 Numerical and symbolic validation

The experimental package contains five independent validation layers.

Exact ReLU recursion. The six inputs and regression windows are those in the public implementation of [4]. The depth-30 calculation generates Table 2; a depth-2000 all-component run and independent stable collision-sector extrapolations verify the coefficients in Table 1. Single-input tests compare all five tensors to the exact polynomials (37).

Orthant sources. Independent Gaussian moments, half-normal singular limits, bivariate angular reductions, and the four-dimensional Plackett integral are tested separately. The deterministic recursion stores a raw quadrature error diagnostic at every layer.

Low-rank vertices. The symbolic Gaussian engine verifies normalized chi-square moments and cumulants through fifth order. The orthogonal engine verifies beta diagonal moments, the covariance (45), and exact vanishing of centered vertices at $r = n$.

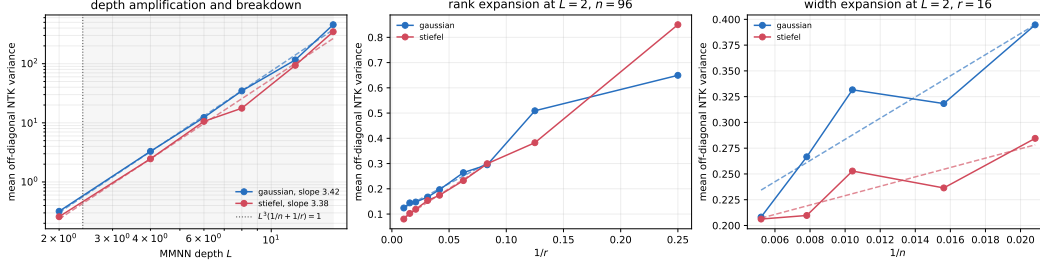


Figure 8: Faithful finite-MMNN NTK variance from the pathwise recursion, with 300 independent networks per point. The shallow rank sweep resolves the $1/r$ contribution (linear-fit $R^2 = 0.986$ for Gaussian and 0.997 for Stiefel factors). Dashed rank/width fits use only points satisfying $L^3(1/n + 1/r) < 1$; the dotted line in the depth panel marks the boundary of that perturbative criterion. The Gaussian and Stiefel ensembles agree in mean metric but differ at finite rank because only the former retains the Wishart radial vertex. The plotted scalar variance is an A/B -sector contraction; it validates the cubic covariance sector and the rank amplitude, not an independent fit of the V, D, F powers.

Pathwise MMNN NTK. For random multi-output, multiblock MMNNs, the recursion (5) agrees to machine precision with an independent explicit propagation of every scalar parameter tangent. A finite-network sweep compares Gaussian frozen factors with their Stiefel-whitened surrogate over width, rank, and depth. It also identifies the nonperturbative region where L^3/r is not small and empirical NTK variances become large and nonmonotone.

Quadratic-scaling concatenation. For the explicit Benigni–Paquette population law with $\gamma_1 = 0.8$, $\gamma_2 = 0.5$, and $\nu = \delta_1$, a deterministic solver evaluates (70) for $L = 1, 2, 4, 8$. It independently verifies (75) against a finite population-law perturbation; the two curves in Figure 4 are indistinguishable at the plotted resolution.

NQF resummation and ignition. An independent implementation forms every parameter gradient $2\mathcal{A}_\mu W$ and verifies (86) against its explicit Gram matrix. It differentiates the closed logistic and matrix-Riccati solutions and compares them with their respective vector fields. In the orthogonal-feature experiment the full numerical eigendecomposition agrees with (96) to machine precision. Finally, Gaussian Wishart and Haar–Stiefel seeds are sampled directly and their ignition-clock means and variances are compared with the digamma and trigamma formulas in Theorem 10.5; see Figure 5.

Spectral resummation. The Dyson density is compared with direct GOE diagonalization, while trace moments are computed independently by noncrossing partitions. Through order eight, simulation means lie within their Monte Carlo errors of the free moment predictions; see Figure 6.

14 Discussion

The exact answer to the motivating question is now sharp. The values printed beside the finite-depth tensor plots are not exact exponents. They are noisy finite-window slopes. The exact critical powers are 1, 2, 2, 3, 3, and the closed coefficients in Theorem 5.2 show that all plotted components attain them. The finite-window variation is physical: critical correlations have a logarithmic drift, and ReLU collision Hessians amplify transverse modes.

For MMNNs, the correct rank calculus alternates width and component indices. The Gaussian Gram cumulant theorem makes every rank vertex explicit; a QR decomposition permits Weingarten evaluation of its orientation without discarding its radial part. This is a more faithful statement than saying that rank merely replaces $1/n$ by $1/r$, or that a Gaussian frozen factor is a projector.

In quadratic sample scaling, the Benigni–Paquette law is the appropriate bulk base. The exact tangent concatenation turns depth into a block population-covariance problem: diagonal blocks carry the transported two-layer laws and off-diagonal blocks carry cross-depth correlations. Under decoupling,

the homogeneous spectrum is the closed MP law (70); without decoupling, the full block covariance must be solved rather than replaced by an unsupported free sum.

The NQF connection adds a training-time resummation to this initialization theory. Its structure matrix is a two-leg Landau vertex and its dynamic Gram matrix is a sufficient order parameter. In the commuting orthogonal sector, mode ignition is exactly NTK-eigenvalue ignition, and low rank enters through an explicit chi-square or beta clock rather than an unspecified finite-size error. This simplification is substantial but sharply delimited: a right-factor-only MMNN block is generally linear, a bilinear pair is locally quadratic, and a generic deeper nonlinear network generates higher-valence vertices that do not close on M .

Spectrally, tensor covariance and random-matrix resummation have distinct jobs. Diagrams compute the covariance and higher cumulants; the Dyson equation computes density and overlaps. A Wigner or Haar law is justified only after checking the resulting covariance sector. This distinction is also necessary for early stopping: eigenvector randomization, spectral lifting, and positivity can compete, so a universal direction of the shift cannot be inferred from a perturbation norm alone.

Important open problems remain. A complete nonperturbative fixed-rank, large-depth law for the product Jacobians in (5) is beyond the double $1/n, 1/r$ expansion. Training-time feature evolution requires the corresponding dNTK and ddNTK collision sectors outside the NQF-resummed subsector. A tractable next step is to compute the first quartic correction to (85), classified by the integer partitions of four, and to propagate its new fourth-moment order parameter through the exact MMNN tangent concatenation. Another is to prove quadratic-form concentration uniformly along the NQF flow, which would upgrade (105) from a pointwise conditional law to a dynamic spectral theorem. Finally, positivity or other factor constraints require cone-projected gradients rather than the Euclidean MMNN tangent metric treated here.

A Proof of the pathwise MMNN recursion

Proof of Theorem 2.1. Partition the parameter vector into the new block parameters (A_ℓ, c_ℓ) and all earlier parameters. Derivatives with respect to the new block satisfy

$$\frac{\partial h_\ell^a(x)}{\partial A_\ell^{b\mu}} = \frac{1}{\sqrt{n_\ell}} \delta_{ab} \phi(z_\ell^\mu(x)), \quad (128)$$

$$\frac{\partial h_\ell^a(x)}{\partial c_\ell^b} = \delta_{ab}. \quad (129)$$

Their contraction at x, y is exactly $q_\ell(x, y)I$. For every earlier parameter ϑ , the chain rule gives

$$\partial_\vartheta h_\ell(x) = J_\ell(x) \partial_\vartheta h_{\ell-1}(x). \quad (130)$$

Contracting the earlier-parameter derivatives at x, y gives the transported term in (5). The two parameter sets are disjoint, so there is no cross term. $\square$

For (7), write $M_{iu} = r^{-1/2} \sum_a W_{ia} A_{au}$. Then $\partial M_{iu} / \partial A_{av} = r^{-1/2} W_{ia} \delta_{uv}$, and direct contraction proves the identity.

B Proofs of the low-rank moment formulas

Proof of Theorem 7.1. Expand

$$\prod_{t=1}^k G_{i_t j_t} = r^{-k} \sum_{a_1, \dots, a_k} \prod_{t=1}^k X_{i_t a_t} X_{j_t a_t}. \quad (131)$$

Isserlis' theorem sums over $\pi \in \mathcal{P}_2(2k)$. A paired Gaussian contraction enforces equality of its row and column indices, producing δ_I^π . The column indices were already identified along τ ; after adding π , one free column label remains per component of $\pi \vee \tau$. Summing them produces $r^{c(\pi \vee \tau)}$, proving (38).

For the cumulant, the moment–cumulant relation groups Wick pairings according to the connected components induced on the k quadratic vertices. Möbius inversion cancels every disconnected grouping and retains precisely $c(\pi \vee \tau) = 1$. Its column factor is r , which together with r^{-k} gives r^{1-k} . $\square$

Proof of Theorem 7.2. Complete U to a Haar orthogonal matrix O . For $P_{ij} = \sum_{a=1}^r O_{ia} O_{ja}$, the orthogonal Weingarten formula gives

$$\mathbb{E} \prod_{s=1}^{2k} O_{I_s A_s} = \sum_{\pi, \sigma \in \mathcal{P}_2(2k)} \delta_I^\pi \delta_A^\sigma \text{Wg}_n^O(\pi, \sigma). \quad (132)$$

The projector definition identifies column indices along τ . Summing over the first r columns gives $r^{c(\sigma \vee \tau)}$. Multiplication by $(n/r)^k$ proves (44). Joint cumulants are the standard Möbius polynomial in these exact moments. If $r = n$, $UU^\top = I$, proving the endpoint statement pathwise. $\square$

C Deterministic orthant evaluation

For a centered Gaussian $X \sim \mathcal{N}(0, K)$, define the truncated moment

$$I_p(K) = \mathbb{E} \left[\prod_i X_i^{p_i} \mathbf{1}_{X_i > 0} \right]. \quad (133)$$

Choose i with $p_i > 0$, reduce p_i by one, and integrate the Gaussian score identity by parts. The result is the Tallis recursion

$$I_p(K) = \sum_j K_{ij} [B_j(p - e_i; K) + (p_j - \delta_{ij}) I_{p - e_i - e_j}(K)], \quad (134)$$

where B_j is the density of X_j at zero times the corresponding conditional truncated moment. Recursion terminates at an orthant probability. In dimensions $d \leq 3$, these are the classical arcsine formulas. For $d = 4$, put $R(t) = I + t(R - I)$. Plackett's identity expresses $d \Pr_{R(t)}(X > 0) / dt$ as a sum of bivariate zero densities times conditional bivariate orthant probabilities, so only a deterministic one-dimensional integral remains. Singular rank one and two cases are obtained before recursion by half-normal and polar integration. This proves that every source in Section 4 is deterministically evaluable without network Monte Carlo.

D Critical asymptotics and collision-sector proof

D.1 Angle and NTK logarithm

The critical ReLU correlation map in angle variables is

$$\cos \theta^+ = \frac{\sin \theta + (\pi - \theta) \cos \theta}{\pi}. \quad (135)$$

Series inversion gives

$$\theta^+ = \theta - \frac{\theta^2}{3\pi} - \frac{\theta^3}{18\pi^2} + O(\theta^4). \quad (136)$$

Set $u_L = 1/\theta_L$. Equation (136) implies

$$u_{L+1} - u_L = \frac{1}{3\pi} + \frac{1}{6\pi^2 u_L} + O(u_L^{-2}). \quad (137)$$

Standard asymptotics for parabolic iterations yield

$$u_L = \frac{L}{3\pi} + \frac{\log L}{2\pi} + C_{ab} + o(1), \quad (138)$$

which proves (28). Substituting $2\dot{Q} = 1 - \theta/\pi$ and $Q = s_a s_b / 2 + O(L^{-2})$ in (19) gives (29)–(30) by a variation-of-constants calculation.

D.2 Collision generators

After division by the external scale $s_a s_b s_c s_d$, the leading collision problem is universal. On a symmetric pair variable X_{ab} , the ReLU Hessian transport has

$$(\mathbf{R}X)_{ab} = X_{ab} + \frac{1}{L} \mathcal{K}_{ab} X + o(L^{-1}), \quad (139)$$

where $\mathcal{K}_{aa} = 0$ and, for $a \neq b$,

$$\mathcal{K}_{ab} X = -3X_{ab} + \frac{3}{2}(X_{aa} + X_{bb}). \quad (140)$$

The gate transport contributes $-3\delta_{ab}/L$. The distributional indicator Hessian has singular transverse entries, but its contraction with the full collision state is finite. Direct substitution of (136) into (22)–(25), retaining the pair-diagonal transverse states generated by (140), gives the following normalized source/damping table:

tensor	power p	order- L^{p-1} source	damping
V	1	5	0
D	2	3	$3\delta_{cd}$
F	2	$2\bar{h}_{bd}$	pair generator
A	3	see the five collision classes below	$3(\delta_{ab} + \delta_{cd})$
B	3	$2\bar{h}_{ac}\bar{h}_{bd}$	$3(\delta_{ab} + \delta_{cd})$

Here $\bar{h}_{ij} = h_{ij}/(s_i s_j)$. The A source depends on the full equality partition of its four external indices, not merely on the number of off-diagonal transported pairs. In exact rational form its five cases are

$$S_{abcd}^A = \begin{cases} 7/4, & \delta_{ab} + \delta_{cd} = 0, \\ 47/40, & \delta_{ab} + \delta_{cd} = 1, \\ 227/320, & \delta_{ab} + \delta_{cd} = 2, \{a, b\} = \{c, d\}, \\ 447/640, & \delta_{ab} + \delta_{cd} = 2, |\{a, b\} \cap \{c, d\}| = 1, \\ 111/160, & \delta_{ab} + \delta_{cd} = 2, \{a, b\} \cap \{c, d\} = \emptyset. \end{cases} \quad (141)$$

For completeness, these constants can be audited term by term. Let $\mathcal{S}_L^A$ denote the right-hand side of (25) with its last, transported- A term removed. After division by $s_a s_b s_c s_d L^2$, the four remaining terms have the following limits; the entry “0” means $o(L^2)$.

collision class	$C_{\Omega, \Omega}$	$\frac{1}{4} \text{OVO}$	$\dot{Q}_{cd} \text{O}_{ab} D$	$\dot{Q}_{ab} \text{O}_{cd} D$	total
both pairs diagonal	1/4	0	3/4	3/4	7/4
exactly one off diagonal	1/16	0	3/10	13/16	47/40
same off-diagonal pair	1/64	0	111/320	111/320	227/320
off-diagonal, one shared label	1/64	0	437/1280	437/1280	447/640
four distinct labels	1/64	0	217/640	217/640	111/160

If the unique off-diagonal pair is the first rather than the second pair, the two D -columns in the second row are interchanged.

To obtain the table without a singular limiting covariance, first use (136) to write, uniformly over the finitely many labels in the component,

$$\frac{K_{ij,L}}{s_i s_j} = 1 - \frac{9\pi^2}{2L^2} \delta_{ij} + O\left(\frac{\log L}{L^3}\right), \quad (142)$$

$$2\dot{Q}_{ij,L} = 1 - \frac{3}{L} \delta_{ij} + O\left(\frac{\log L}{L^2}\right). \quad (143)$$

Represent the standardized Gaussian variable for each distinct label as a common normal plus independent residuals of size $3\pi/(\sqrt{2}L)$, rescale the common normal by L in the gate-boundary layer, and apply (134). The half-normal integrals give the first column of the table. Substitution of the already solved V and D collision states into the explicit Hessians gives the next three columns. All remainders are dominated by a Gaussian polynomial and are uniform over a fixed finite input set, so dominated convergence justifies the termwise limits. This is also a direct finite calculation of every rational number in (141), rather than a fit to the numerical recursion.

The transverse terms change the naive off-diagonal D , A sources; omitting them is the leading-mode error discussed in the main text.

We use the following elementary discrete transport lemma to turn the formal balance into an asymptotic statement. If a finite-dimensional collision state satisfies

$$X_{L+1} = \left(I + \frac{\mathcal{K}}{L} + o(L^{-1})\right) X_L + L^{p-1}(S + o(1)) \quad (144)$$

and every eigenvalue of $\mathcal{K}$ has real part strictly below p , then

$$L^{-p} X_L \longrightarrow (pI - \mathcal{K})^{-1} S. \quad (145)$$

Indeed, after setting $Y_L = L^{-p} X_L$, expansion of $(L/(L+1))^p$ gives $Y_{L+1} - Y_L = L^{-1}[(\mathcal{K} - pI)Y_L + S + o(1)]$. Variation of constants (or a Jordan decomposition followed by summation) shows that the stable recursion converges to its unique fixed point. The collision matrices here have eigenvalues among 0, -3 , -6 , so the hypothesis holds for $p = 1, 2, 3$.

In a scalar sector with transport $1 - g/L + o(L^{-1})$, the lemma gives $(p+g)t = \text{source}$. Applying it to the table proves (32)–(36). The F pair generator annihilates the vector proportional to $\bar{h}_{bd}$, giving $f_{abcd} = s_a s_c h_{bd}$. Every displayed source and denominator is positive, proving non-cancellation.

E Dyson and free-moment derivation

For a Gaussian correction with covariance operator $\mathcal{S}$, planar rainbow diagrams for the averaged resolvent close under insertion of one covariance line. Their Schwinger–Dyson resummation is

exactly (110). Under GOE covariance, $\mathcal{S}[R] = \sigma^2 m^{-1} \text{Tr}(R)I$, so the diagonal entries give (112)–(113). The Stieltjes inversion formula proves (111) and (114). Since $M_{ii}(z) = \int (\lambda - z)^{-1} d\mu_i(\lambda)$, the spectral measure μ_i has atoms $|\langle e_i, u_j \rangle|^2$ at the sample eigenvalues. Dividing its local mass by the local eigenvalue count $m\rho(\lambda)$ proves (115).

The semicircle law has the single nonzero free cumulant $r_2 = \sigma^2$. Asymptotic freeness of deterministic diagonal matrices and GOE matrices makes free cumulants additive, proving (116). The moment–cumulant relation

$$m_k = \sum_{\pi \in NC(k)} \prod_{B \in \pi} r_{|B|} \quad (146)$$

then gives every trace moment. For the PSD model (124), solve the same equation for $B = \text{diag}(\sqrt{\eta}) + \sigma W$, push the local measures forward by $s \mapsto s^2$, and insert them into (123).

F Power-law asymptotics

With $q = 1 + \alpha$, substitution $u = 2Pt\eta_1 x^{-q}$ in the continuum approximation gives

$$\mathcal{L}_{\text{bias}}(t) = \frac{\eta_1 \Gamma(\alpha/(1+\alpha))}{2(1+\alpha)} (2Pt\eta_1)^{-\alpha/(1+\alpha)} (1 + o(1)), \quad (147)$$

$$\mathcal{L}_{\text{var}}(t) = \frac{\Gamma(\alpha/(1+\alpha))}{2P\beta} (2Pt\eta_1)^{1/(1+\alpha)} (1 + o(1)). \quad (148)$$

Differentiation gives (121). Weyl’s inequality gives eigenvalue displacement at most s , and Davis–Kahan bounds the angle of a simple population eigenvector by $s/(g_j - s)$. Substitution of the power-law gap yields (126). Requiring the cutoff learned by time t_0^* to remain below J_{mix} gives (127).

G Reproducibility

The principal scripts are:

- `run_exact_relu_tensor_recursions.py`: deterministic finite-depth tensors and orthant diagnostics;
- `relu_tensor_asymptotics.py`: exact coefficients and componentwise convergence;
- `certify_relu_collision_sources.py`: termwise rational certificate for all five A -source collision classes;
- `compare_paper_depth_exponents.py`: identical-window slope comparison;
- `exact_mmnn_ntk_recursion.py`: pathwise MMNN NTK;
- `exact_mmnn_gram_wick.py` and `exact_orthogonal_weingarten.py`: all-valence rank vertices;
- `benigni_paquette_mmnn_spectrum.py`: concatenated quadratic-scaling spectrum and exact MP linear response;
- `nqf_mmnn_mode_ignition.py`: exact NQF NTK spectrum, isotropic Riccati resummation, and Wishart–Stiefel ignition clocks;
- `deformed_wigner_dyson.py`: moments, density, and overlaps;
- `run_dyson_powerlaw_early_stopping.py`: PSD spectral learning curves;
- `run_mmnn_ntk_variance_scaling.py`: faithful finite-MMNN width/rank/depth sweep.

Automated tests verify the pathwise recursion, single-input polynomials, orthant moments, componentwise coefficients, chi-square and beta moments, connected cumulants, semicircle moments, Dyson overlap sum rules, and power-law spectral costs. The NQF tests additionally verify its NTK against explicit parameter gradients, the complete orthogonal-mode spectrum, the Riccati vector field, the chi-square/beta log moments, and full-rank Stiefel clock cancellation.

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